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REVIEW

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Safeguarding the skies: the rise of machine learning approaches for cyberattack detection in unmanned aerial systems

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Abstract

Unmanned Aerial Systems (UAS) have become increasingly integral to industries such as agriculture, surveillance, transportation, and entertainment, raising significant concerns about their vulnerability to cyberthreats. Concurrently, cyberattacks have grown drastically in both frequency and complexity. In response, researchers have increasingly turned to Machine Learning (ML) techniques to improve UAS security, though accurately detecting attacks with low computational demands remains a challenge. This Systematic Literature Review (SLR), following PRISMA guidelines, examines ML-based methods for detecting cyberattacks on UAS, analyzing 101 articles published between January 2019 and May 2025. The review summarizes findings on types of feature data, preprocessing techniques, learning approaches, evaluation metrics, and detection designs used in ML-driven cyberattack detection for UAS. Unlike previous reviews, this study addresses the practical implementation of these approaches in UAS. It also highlights key techniques from the literature to help researchers build robust ML models. Furthermore, the SLR outlines the limitations and challenges of ML-based detection methods, including issues related to datasets, evaluation, model complexity, interpretability, and experimental aspects during training and validation. Finally, the review proposes future research directions to advance the field.

Keywords Unmanned aerial systems, Drones, Unmanned aerial vehicles, Attack detection, Intrusion detection systems, Machine learning

Introduction

Over the last two decades, Unmanned Aerial Systems (UAS), known as drone systems, have gained popularity, and the missions assigned to them have evolved rapidly in all domains, ranging from simple food delivery to complex military operations, as shown in Fig. 1. Additionally, UAS have emerged as a crucial component in the information technology infrastructure of smart cities which rely on networked sensors and relays to collect, exchange and process data (Ullah et al. 2020). To ensure trustworthy digital services (e.g., data transmission, surveillance), UAS in smart cities need to be safe, secure, and reliable. Failing that, these flying objects could become de facto 'invisible killers' (Syed et al. 2021) in the event of a cybersecurity failure. As the security of communication links and sensor input data is critical to maintaining mission integrity and operational safety (Guerber et al. 2021), a UAS cybersecurity failure can sometimes result in a disastrous event. For example, kamikaze drones, which can be deployed in swarms to launch attacks on adversaries on a battlefield theater (Husodo et al. 2020), could be misled to target friendly troops due to simple Global Positioning Systems (GPS)

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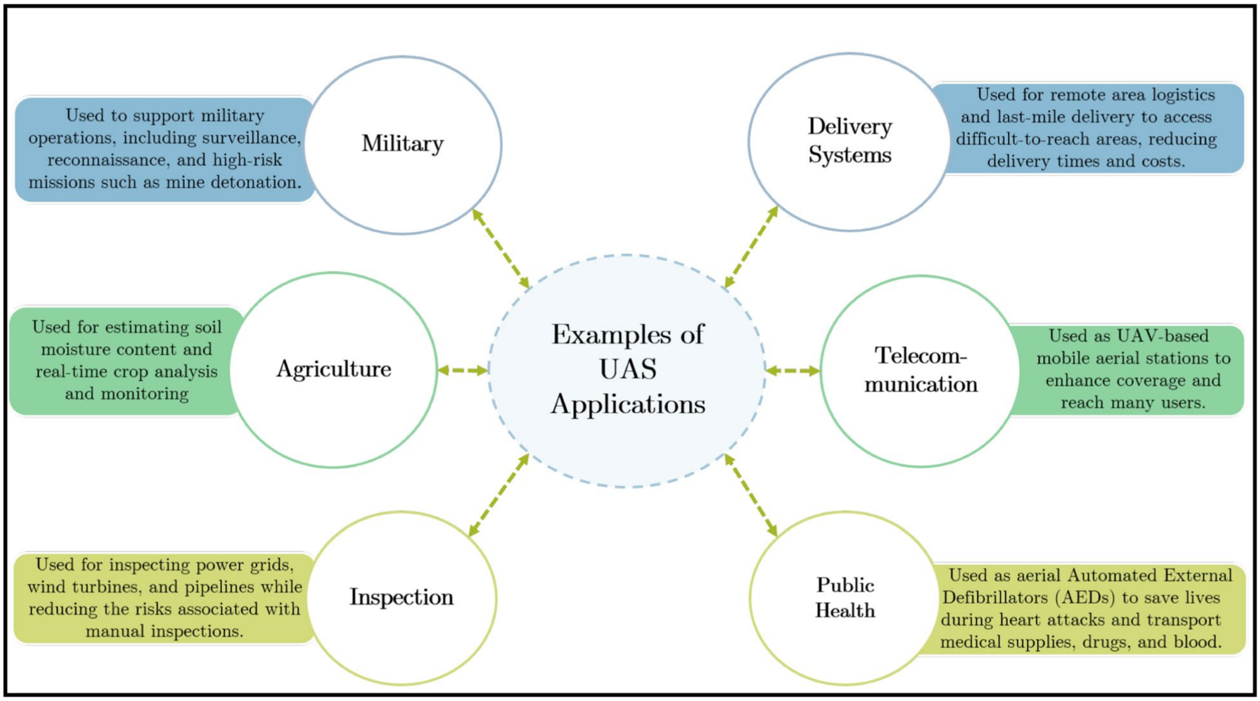

Fig. 1 Examples of UAS applications

spoofing. In fact, a proliferation of cyberattacks targeting UAS has been registered recently, posing a risk not only to their systems' users but also to flight and public safety. Examples of recent incidents include GPS jamming that occurred late in 2018, causing 46 drones to fall from the sky during a Hong Kong show, leading to a loss of approximately one million Hong Kong dollars. In 2020, another incident occurred in China at the lantern festival (Nayfeh et al. 2023), resulting in drone crashes during a light display.

These incidents have drawn significant attention from academia and industry, underscoring the need for advanced cybersecurity detection and protection solutions. For instance, Lockheed Martin has developed and tested a prototype solution to detect hardware-level cyberattacks against UAVs, with minimal processor impact (LockheedMartin 2025a). Given that many of Lockheed Martin's UAS, such as the Indago 4 quadcopter, already support on-edge Machine Learning (ML) for reconnaissance and surveillance (LockheedMartin 2025b), it is likely that this prototype is also ML-based. Furthermore, while DJI[1] does not explicitly promote a proprietary ML-based IDS or security module in its products (DJI 2025a), the drone manufacturer offers powerful companion computers like Manifold 2 (DJI 2025b) with GPU capabilities, enabling developers to implement custom ML models for security purposes, including potential IDS applications.

UAS components and architecture

Typically, a UAS consists of an Unmanned Aerial Vehicle (UAV), a Ground Control System (GCS), and a data link. The UAV is considered a Cyber Physical System (CPS) (Leccadito et al. 2018), which incorporates networking as well as computational capabilities that ensure the execution of physical processes, such as relying on the input of sensors to operate actuators according to algorithms. The embedded sensors in the UAV include the Inertial Measurement Unit (IMU), barometers, and GPS receiver, among others. Depending on its application scope, the UAV can be equipped with payloads such as cameras, missiles, and radars. The GCS can be a hardware system installed in a fixed position or mounted on a vehicle, or simply a software solution running on a personal computer or a mobile device, as shown in Fig. 2. For the hardware configuration, the GCS can comprise a cockpit to enable operators to send mission commands to the UAV, in return this latter sends telemetry, i.e. UAV's position, altitude, speed, orientation, battery status, and other relevant parameters.

In case of Visual Line-Of-Sight (VLOS) between the UAV and the GCS, the data link uses direct radio waves (Yaacoub et al. 2020), such as Radio Frequency, Wi-Fi, and Zigbee, among others. If the UAV is Beyond Visual

[1] One of the leading manufacturers of commercial drones.

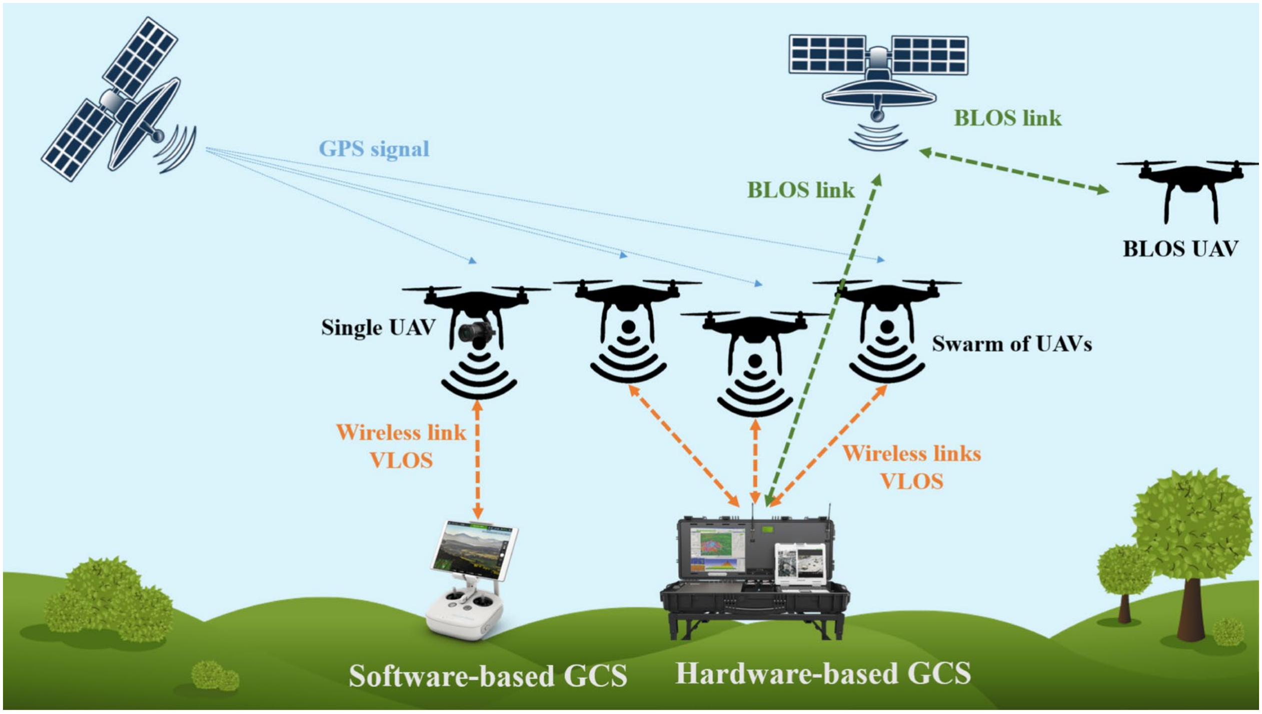

Fig. 2 UAS components and architecture

Line-Of-Sight (BLOS), it is controlled by the GCS via satellite (Yaacoub et al. 2020) or cellular technology, as highlighted in Fig. 2.

Regarding their operations, UAS can consist of a single vehicle or a swarm of UAVs, in which case researchers use the term Internet of Drones (IoD). Yahuza et al. (2021) defined IoD as a decentralized control network architecture that enables multiple interconnected UAVs to use airspace while providing them with navigation services to locate each other. In addition, UAVs can be remotely controlled via GCS, in which case they are referred to as Remotely Piloted Aircraft Systems (RPAS) (International Civil Aviation Organization 2011), or they can fly autonomously without the need for the intervention of a drone operator.

Research problem

Despite being perceived as highly sophisticated systems, UAS are, in fact, poorly protected against intruders and lack inbuilt protection (Nassi et al. 2021), leaving their operations vulnerable to easy exploits (Vasconcelos et al. 2016; Westerlund and Asif 2019). Recently, ML has been used to tackle cybersecurity challenges in UAS, including vulnerability assessment (Veerappan et al. 2021), anomaly detection (Alzahrani 2024), and forensics (Abdulrahman Debas et al. 2024). Indeed, detecting attacks on drone systems using ML approaches is critical but challenging due to numerous technical and operational complexities. While these approaches require specialized ML techniques, there remains a notable lack of comprehensive reviews focused on attack detection methods specifically designed for drone systems. Bridging this gap is essential for advancing research and fostering the development of more robust and reliable intrusion detection solutions for UAS.

Related works

Several reviews and surveys addressed the detection of cyberattacks. Kong (2021) surveyed the countermeasures of cyberattacks targeting UAVs. The countermeasures have been classified into prevention, detection, and mitigation solutions. Six types of attacks have been considered, namely on-board system attacks, channel jamming, message interception, message spoofing, and message injection. However, the study focused on works outside the realm of ML.

Bangui and Buhnova (2021) listed, without going into detail, ML works related to Intrusion Detection Systems (IDS) for Vehicular Ad-Hoc Networks (VANET), and particularly UAV-aided networks. The study elaborated on detection mode, i.e. signature-based and anomaly-based, as well as deployment of IDS, on-board detection, collaborative detection and detection in the cloud.

Syed et al. (2021) provided a comprehensive review of security solutions based on ML, blockchain, and watermarking to secure data transmission in UAS. The survey discussed UAV applications, their use cases, their requirements, their challenges, and their contribution to secure Internet of Things (IoT) systems. The surveyed papers included articles involving securing UAV collaboration, such as solutions related to UAV collision avoidance methods, namely detection of objects with UAV imagery, sound-based amateur drone detection, and identification of drones. Nevertheless, the authors focused on object detection to ensure safe collaboration between drones and did not thoroughly investigate solutions to the detection of intrusion in UAS networks.

Tsao et al. (2022) provided a taxonomy of security threats in relation to threat vectors targeting each communicating part of a UAS: terminals and GCS, GCS and backbone UAV, UAV-UAV, UAV requesting to join the Flying Ad-hoc Networks (FANET), and cloud services connections. The study provided solutions to mitigate FANET security threats and proposed a match between security requirements, security threats, and security solutions. However, the surveyed works were limited to the lower layers of the network.

Ferrag et al. (2022) reviewed IDS for securing Agriculture 4.0 against cyber threats. The paper analyzed cybersecurity risks and evaluation metrics for IDS performance in this domain. It classified ML and Deep Learning (DL)-based IDS across emerging technologies like Cloud computing, Fog/Edge computing, IoT, Smart Grids, and drones. It also details best practices, public datasets, and implementation frameworks used in IDS evaluations. While it provides valuable insights on IDS for drone-enabled agriculture, it lacks discussion on how to deploy these approaches.

Alsumayt et al. (2024) reviewed ML techniques utilized to detect Denial-of-Service (DoS) attacks in IoD networks. They compared the results of relevant works and highlighted their limitations. In addition, the paper presented vulnerabilities as well as threats to UAVs, focused on DoS attacks against drone networks, and proposed a classification of these attacks. Branco et al. (2025) analyzed cyberattacks on UAVs, focusing on vulnerabilities in components such as GCS and communication protocols. The study explored various attack methods, and reviewed existing penetration testing methodologies for UAVs, detailing tools and techniques used to execute these attacks. It offers an accessible analysis of drone vulnerabilities to help improve cybersecurity measures for UAVs.

Alzubaidi (2025) surveyed recent advances in the detection of malware in UAVs, focusing on ML-based approaches. The study provides a comprehensive review of detection methods, analyzing common datasets, feature selection, and extraction techniques critical to developing intrusion detection frameworks. Furthermore, it introduces a novel taxonomy for malware detection approaches, and identifies limitations in existing studies.

The analysis of related works reveals that some reviews explored non-ML techniques to secure UAVs, while others lacked sufficient technical detail on the ML approaches employed in the surveyed studies. Furthermore, certain surveys examined intrusion detection in general but omitted discussion of practical deployment of these systems, whereas others concentrated exclusively on specific cyberattacks targeting UAVs, such as DoS or malware. Although there are some comprehensive and valuable studies, they are not specifically focused on UAVs.

We further compare related works based on the following criteria and present the results in Table 1.

- SLR: Indicates whether the review adopts the structure and the methodology of a Systematic Literature Review (SLR).

Table 1 Examples of attack techniques used against UAS

Related work	SLR	UAS context	ML approaches			Deployment
			Preprocessing	Learning paradigm	Evaluation	
Kong, (2021)	⊠	✓	⊠	⊠	⊠	⊠
Bangui & Buhnova, (2021)	⊠	✓	⊠	✓	⊠	⊠
Syed et al., (2021)	⊠	✓	⊠	✓	⊠	⊠
Tsao et al., (2022)	⊠	✓	⊠	⊠	⊠	⊠
Ferrag et al., (2022)	⊠	⊠	✓	✓	✓	⊠
Alsumayt et al., (2024)	⊠	✓	⊠	✓	✓	⊠
Branco et al., (2025)	✓	✓	⊠	⊠	⊠	⊠
Alzubaidi, (2025)	⊠	✓	✓	✓	✓	⊠
Our study	✓	✓	✓	✓	✓	✓

- UAS context: Specifies whether the review thoroughly discusses studies applied to the UAS domain.
- ML approaches: Specifies whether the reviewed studies include ML approaches, detailing the extent to which these approaches are analyzed, including preprocessing techniques, learning paradigms, and evaluation methods.
- Deployment: Indicates whether the review examines the practical deployment of proposed approaches in the studies, including their implementation on hardware assets.

Contribution

The contributions of this SLR are as follows:

- We provide a comprehensive analysis of ML-based approaches for detecting multiple attacks on UAS, detailing the entire process, including preprocessing, learning, and evaluation.
- We thoroughly discuss the limitations and challenges of these approaches.
- We examine deployment aspects of ML-based attack detection in the context of UAS.

To the best of our knowledge, no prior work has addressed the deployment of ML approaches on UAS. In addition, our study serves as a valuable starting point for new researchers aiming to contribute to the field of drone cyberattack detection. By reviewing ML approaches, it provides the scientific community with comprehensive insights on how to develop lightweight and robust approaches suitable for use on UAS.

Paper organization

The remainder of this paper is organized as follows. In the next section, a background knowledge of cyberattack detection is provided. The research methodology including the selection process, the research questions(RQs), the inclusion and exclusion criteria, and an overview of selected articles is presented in Section (Research methodology). Section (Review of ML approaches) provides a detailed and critical review of ML approaches employed for detecting cyberattacks on drone systems. The limitations and challenges of the reviewed studies are discussed in Section (Limitations and challenges). Lastly, this systematic review study concludes in Section (Conclusion), and provides potential future research directions.

Background

A cyberattack is a deliberate action carried out by a malicious threat agent with the aim of compromising the security of systems, networks, or data. Cyberattacks against a UAS can target any component of the system and rely on different types of techniques, such as GPS spoofing, DoS, and Man-in-the-Middle (MITM) (Yahuza et al. 2021). Some attack techniques, their description, and exploit examples are provided in Table 2. Researchers have suggested multiple mechanisms for identifying cyberattacks. This section explores the landscape of cyberattack detection, with a focus on IDS, the role of ML in cyberattack detection, and the challenges associated with the deployment of these technologies in different environments.

Cyberattack detection

In UAS, cyberattack detection is crucial to ensure the safety and security of both the vehicle and its mission. Implementing advanced cyberattack detection mechanisms helps identify and respond to these threats in real time. As these attacks often aim to gain control or disrupt the UAS's operations, the emphasis shifts towards intrusion detection.

Intrusion Detection Systems

Intrusion is unauthorized access by an entity to resources that compromises its confidentiality, integrity, and availability. The decision-making process for detecting attacks or intrusions within a UAS can be based on different information gathering sources (Choudhary et al. 2018), namely UAV components, sensors, communication links, and GCS. Network IDS (NIDS) are systems that can make the difference between normal and malicious traffic by profiling normal and abnormal behavior or by relying on specific rules or specific patterns within the traffic, known as signature. They constitute a second layer of protection and can be deployed along with other security mechanisms, such as encryption and authentication (Ferrag et al. 2020).

Deployment of detection solutions

IDS specifically designed for UAS require minimal computational resources to offer real-time detection while saving energy (Wang et al. 2021). It can be deployed on-board or off-board, depending on the lightweightness of the proposed solution. Researchers can choose to deploy it on-board. In this case, all processing is done within the vehicle sharing resources, which are usually used for navigation and collision avoidance. Otherwise, they can choose to deploy the solution in the GCS (Basan et al. 2021), or in the cloud due to resource constraints. Some researchers proposed a distributed deployment (Khraisat and Alazab 2021), where UAVs check each other for possible intrusions.

ML for Cyberattack detection

Detecting cyberattacks is a classification problem that can be addressed with ML approaches. An example of solution is anomaly-based intrusion detection where the normal behavior is profiled through ML algorithms, and when implemented, every behavior deviating from the normal state is classified as intrusion. Several

Table 2 Examples of attack techniques used against UAS

Attack technique	Description	Exploit example
GPS spoofing	Deceiving the UAV's GPS sensor by broadcasting fake GPS signals to be substituted for the legitimate signals	Hijacking of DJI Phantom 3 via GPS spoofing (Zheng and Sun 2020)
Data link jamming	Preventing the transmission of data (commands, telemetry and payload data) between the UAV and the GCS	Parrot UAV-GCS data link jamming (Multerer et al. 2017)
Denial of Service (DoS)	Flooding the communication link with a huge amount of traffic leading to denial of access to legitimate users	DoS attack against a Parrot AR UAS (Vasconcelos et al. 2016)
Man-In-The-Middle (MITM)	Taking position between the GCS and UAVs or between UAVs to intercept and alter traffic exchange	Interception of traffic in Parrot AR and CX-10W UAS (Westerlund and Asif 2019)
GPS jamming	Preventing the UAV from receiving the GPS signal by causing interference within the targeted GPS receiver	GPS jamming against a DJI Phantom 4 Pro (Rahman et al. 2021)
False Data Injection (FDI)	Altering, substituting, or modifying images or videos obtained via a UAS to change decision-making	FDI in Ardupilot flight controller system (Chen et al. 2017)
Hardware Trojan (HT)	Trojan initially disabled, waiting for predefined conditions to be met to perform malicious activities	Modification of the UAV motors' rotation speed (Rahman et al. 2020)
Wormhole Attack	Making remote UAVs, in a UAS swarm, believe that they are neighbors by imposing the shortest available route	Exploiting a pair of UAVs to form a wormhole tunnel (Wang et al. 2021)
ADS-B spoofing	Injecting fake ADS-B signals from fictive aircraft to force a UAV into changing its trajectory to a desired direction	Aircraft Target Ghost Inject exploit (McCallie et al. 2011)
Sybil attack	Deceitfully asserting that a UAV possesses multiple identities to gain additional influence	Not available
Identity spoofing	Pretending to be another UAV or GCS by falsifying identity-related information to gain trust	Not available
Command injection	Injecting new flight commands, which are perceived by the UAV's flight control system as legitimate commands	Command injection in a Pixhawk-based UAV (Cammilleri 2015)
Buffer overflow	Manipulating memory operations to cause an overflow in the UAS buffers giving unauthorized access	Exploit against Parrot Bebop UAS (Hooper et al. 2016)
Unauthorized root access	Exploiting UAV-GCS communication protocol vulnerabilities to gain privileged access	Telnet exploitation in Parrot AR UAS (Westerlund and Asif 2019)
Keylogger attacks	Collecting all sensitive information processed by GCS through a Keylogger software	Keylogger found in the GCS of Predator & Reaper UAS (Shachtman 2011)
Malware infection	Enabling an adversary to remotely take control of a UAV after inserting a malicious software	Maldrone infection in Parrot AR drones (Jiang et al. 2020)

classification algorithms are used to detect attacks on UAS following two approaches. The first is binary classification, which considers only two classes, such as normal and malicious traffic. The second is multiclass classification, and consists of adopting more than two classes, hence classifying each instance into several categories, i.e., classes which are in our context attack types. The challenge of these detection systems is to overcome the high false positive rates (Mouatassim e al. 2024).

Research methodology

Employing a research methodology guarantees the validity and reliability of the research findings. In addition, it helps the researcher to systematically plan and structure their research. There are multiple frameworks to consider when conducting an SLR. Popular methodologies include Kitchenham's guideline (Kitchenham et al. 2009), Joanna Briggs Institute (JBI) approach (Moola et al. 2015) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyzes (PRISMA) framework (Rethlefsen et al. 2021). Unlike other methodologies, the latter is a multidisciplinary framework that enhances the transparency of the systematic review process through its comprehensive checklist and flow diagram. For these reasons, we opt for the guidelines and recommendations of the PRISMA framework.

Time period and search keywords

This SLR was conducted from 11/13/2024 to 06/13/2025 on works published from the beginning of January 2019 to the end of May 2025. The research strategy was based on a combination of precise keywords that serve the purpose of the SLR while including potential variation for each keyword. The research was performed on title, abstract, as well as authors' keywords in the Scopus and Web of Science bibliographic databases. The search request was divided into four parts. In the first part, we searched for the term "detection" or "detecting". The second part dealt with the term "attack" or "cyberattack" or "cyber-attack" or "intrusion". The third part was about the terms "drone" or "unmanned aerial vehicle" or

Table 3 Exclusion and Inclusion criteria and their description

Criteria	IC/EC	Description
Inclusion	IC1: Scope	Works dedicated to detection of intrusion or cyberattacks in UAS
	IC2: Technique	Studies that leverage ML as a solution to the research problem
	IC3: Language	Papers written in English
Exclusion	EC1: Purpose	Studies irrelevant to the subject, and not dealing directly with the research problem
	EC2: Scope	Studies not specific to UAS
	EC3: Technique	Detection technique outside the realm of ML
	EC4: Quality	Inconsistent research, no result, general methodology and lack of originality
	EC5: Redundancy	Similar study and results from the same authors

"unmanned aerial system". The last one looked for the terms "artificial intelligence" or "machine learning" or "deep learning" or "neural network".

Inclusion and Exclusion criteria

During the selection process, the studies underwent screening to determine their eligibility for inclusion or exclusion according to the Inclusion Criteria (IC) and Exclusion Criteria (EC) outlined in Table 3. For inclusion in this SLR, studies were required to meet specific criteria: they must be dedicated to the detection of UAS intrusions or cyberattacks (IC1), must employ a ML approach (IC2), and must be written in English (IC3). In contrast, studies were excluded if they did not meet the following criteria: they were not directly related to the UAS cyberattack detection research problem (EC1), were not specific to UAS (EC2), did not provide enough detail about the proposed ML approach (EC3), lacked original results and a specific methodology (EC4), or presented redundant results across multiple publications (EC5). These criteria ensured the selection of relevant, original, and non-redundant studies.

Selection process and tools

The process adopted in our study consists of three phases: identification, screening, and eligibility, as depicted in Fig. 3. First, we identified 680 papers from two bibliographic databases, namely Scopus and Web of Sciences. Then, the initial set of studies was examined for duplicates. After discarding 123 duplicated works, we systematically screened articles by evaluating their titles, abstracts, and author keywords against the inclusion and exclusion criteria. For example, the study titled "User-Agent as a Cyber Intrusion Artifact: Detection of APT Activity Using Minimal Anomalies in User-Agent String Traffic" was excluded (EC1) because it is not relevant to the research problem. This study was identified in bibliographic databases among the initial set of studies because of its use of the abbreviation "UAS" for User-Agent String. Thus, the preliminary assessment allowed the exclusion of 351 papers, most of which focused on drone detection, object detection by drones, or other topics unrelated to the research problem.

Furthermore, three relevant studies that were not found in the databases were identified using a snowball search method and subsequently included in our study. Next, we performed an eligibility evaluation by full-reading 203 studies and verifying inclusion and exclusion criteria. The full-text assessment allowed the exclusion of half of these papers. For instance, the article titled "A Data Normalization Technique for Detecting Cyber Attacks on UAVs" was excluded because it did not employ ML techniques (EC3). Upon review, we determined that its methodology relied on probability theory and statistical methods. Similarly, the article titled "A Machine Learning Approach for Detecting and Classifying Jamming Attacks Against OFDM-Based UAVs" was excluded due to redundancy(EC5), as the results presented in this conference paper were similar to the results included in a journal article published by the same authors, which was already included in our SLR.

Ultimately, 101 studies were retained for inclusion in our review. Throughout the selection process, we employed Rayyan[2], a dedicated online platform for systematic reviews. This tool contributed significantly to simplifying the study selection procedure. It helped in the article screening process, promoting collaborative decision-making, and improving the transparency of our review methodology.

To enhance the transparency and reproducibility of this SLR, we provide access to documentation via a GitHub repository,[3] detailing the rationale for each study's inclusion or exclusion.

Overview of the selected studies

The research problem addressed in this review is gaining in popularity among scholars, as illustrated in Fig. 4. The figure indicates that the number of studies increased

[2] https://www.rayyan.ai/

[3] https://github.com/TaarekMM/SLR/

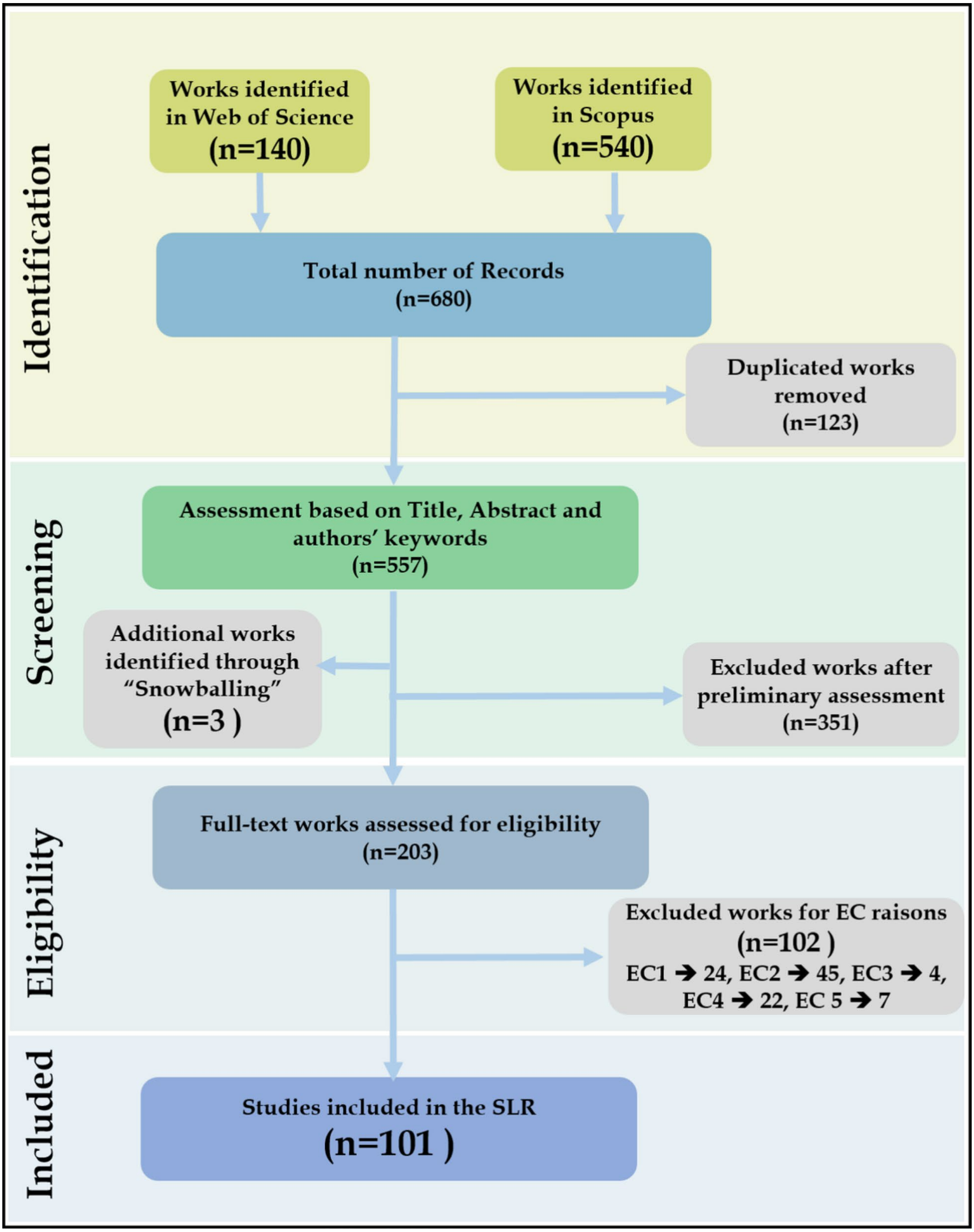

Fig. 3 PRISMA process for identifying studies to be included in our SLR

Table 4 Research questions

RQ1	Approaches	(a) What are the types of feature data used as input to ML models for attack detection?
		(b) How do preprocessing techniques apply to this domain?
		(c) What are the main intelligent paradigms used during the learning phase?
		(d) How did the researchers evaluate their proposed approaches?
		(e) How did the researchers implement their proposed models?
RQ2	Limitations	(a) What are the main limitations of ML approaches in detecting attacks on UAS?
		(b) What are the common challenges faced by ML approaches in detecting attacks on UAS?

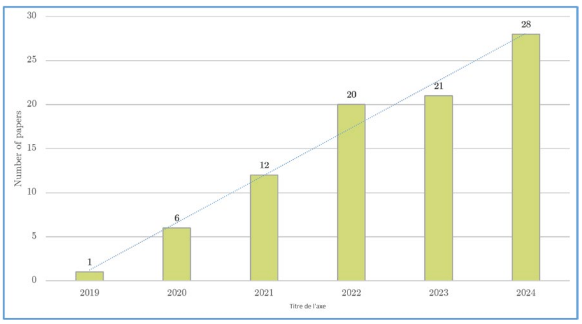

Fig. 4 Distribution of selected studies by time period 2019–2024

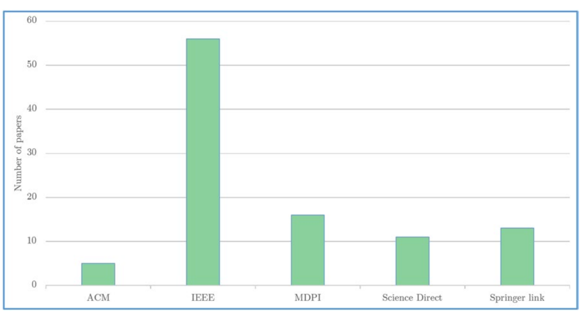

Fig. 5 Distribution of selected studies by publisher

significantly in the past six years (2019-2024), with 80% of reviewed papers published in the last three years. In addition, highly respected and renowned publishers, such as ACM, IEEE, MDPI, Science Direct, and Spring link, are paying great attention to this field, as shown in Fig. 5.

Research Questions

We formulated ideas for analyzing papers and developed specific research questions (RQs) with detailed sub-questions, as presented in Table 4, to guide our investigation. First, we study the ML approaches that have been proposed to detect attacks targeting UAS (RQ1), including the types of data feature (RQ1 (a)), preprocessing techniques (RQ1 (b)), and the learning paradigms used by researchers in the leaning phase (RQ1(c)). We also investigate how scholars evaluated (RQ1(d)) and implemented (RQ1(e)) their proposed solutions. Second, we focus on the main limitations of ML approaches (RQ2(a)) for attack detection, and their corresponding challenges (RQ2(b)).

Review of ML approaches

UAS increasingly rely on ML algorithms to enhance security through attack detection. The process begins with collecting and preprocessing data to ensure quality and consistency. This data is then divided into training data and testing data. ML algorithms are trained on the training data to develop models capable of identifying potential intrusions. Once trained, these models are applied to the testing data where the models are evaluated based on several criteria to assess their effectiveness in detecting attacks. Finally, the most successful models are implemented for real-world attack detection to recognize cyberattacks.

This process provides a framework for analyzing and discussing the approaches proposed in the primary studies retained in our SLR, which we examine in this section to address the sub-questions of **RQ1**. Fig. 6 illustrates the alignment between the subquestions and the general methodology for developing and implementing ML models for attack detection in UAS. Table 6 provides a summary of the primary studies included in our SLR, highlighting the approaches, datasets used, the ML models investigated, the best results achieved, the strengths and limitations of each study.

Types of feature data

The type of feature data provided to ML models for attack detection in UAS is critical to ensure effective detection. These features vary depending on the specific attack types targeted and can be combined to enhance the detection accuracy for specific attacks or to broaden the scope of detection. It includes flight control, sensor data, payload data, internal circuitry and external signal characteristics, network traffic, and system logs, as depicted in Fig. 7.

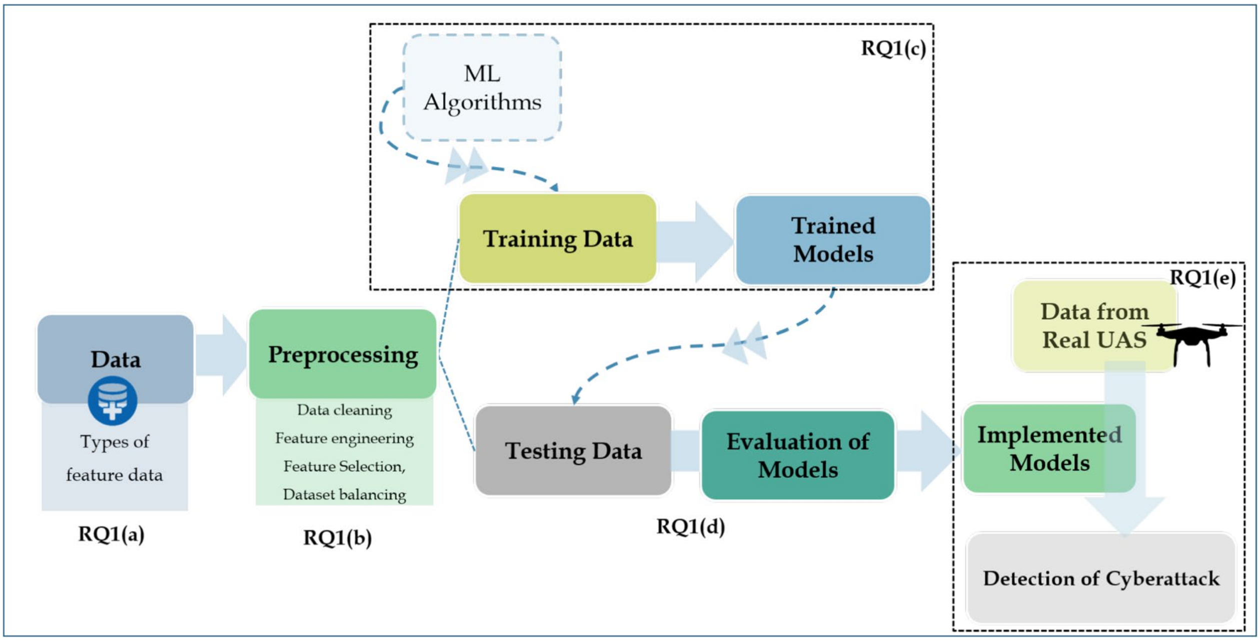

Fig. 6 Methodology of cyberattack detection in UAS

Flight control and sensor Data

Flight control logs and internal sensor data are automatically generated during UAV flights. They can be stored for post-analysis in the flight controller or transmitted via data links to GCS or other UAVs. They include time-stamped data about the UAV's operations, such as telemetry, sensor readings, GPS data, and control inputs.

This type of feature is particularly utilized to detect GPS spoofing attacks. Alzahrani (2024) proposed multi-sensor anomaly detection approach. The author investigated the impact of considering all UAV sensors compared to considering a restricted number of sensors. The author concluded that using the data coming from all sensors leads to better performances than considering a restricted data source. A similar approach was adopted by Wei et al. (2025), where data from heterogeneous sensors, including inertial sensors, barometers, and GPS were fused to improve classification accuracy and robustness. Meanwhile, Feng et al. (2020) proposed an ML approach using only GPS data, i.e. longitude and latitude, and IMU data, i.e. angle velocity and acceleration. Compared with the multi-sensor approach, the latter approach can only detect attacks with behaviors similar to those observed during training.

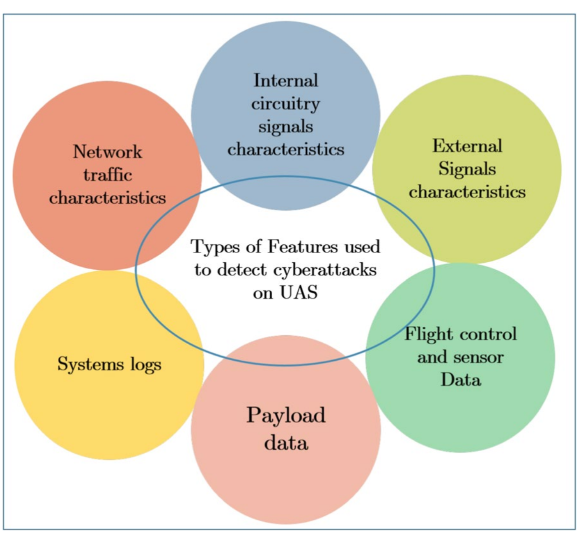

Fig. 7 Types of features feed to ML models for attack detection

Wei et al. (2022) proposed an approach based on control semantics, where rules and relationships, which define how control inputs translate into UAV actions, are examined for attack detection. The features were selected on the basis of the horizontal position control process and the altitude control process. This approach was compared with the method presented in (Feng et al. 2020) and demonstrated its superiority. Wei et al. (2022) proposed the PerDet framework that relies on perception data from complementary sensors, namely accelerometer, gyroscope, magnetometer, GPS and barometer sensors, to establish the UAV's attitude and position estimations through ML and accordingly identify GPS spoofing. A common challenge of these approaches is the generalizability of ML models across different UAV platforms, which have been investigated by Agyapong et al. (2021). The authors compared the performance of models trained on telemetry data from multiple types of UAVs with those trained on data from a single UAV platform, finding that the latter models significantly outperformed the former models.

Flight control and sensors data features can also be used for the detection of GPS spoofing attacks by predicting the UAV path and then comparing this value with the expected path. Wang et al. (2020) proposed predicting the path of the UAV giving IMU information. The predicted positions are compared with the GPS positions received to confirm the attack when a default error is exceeded. This solution may result in a high False Alarm Rate (FAR) in the case of perturbations.

A major advantage of using flight control and sensor data features for UAV cyberattack detection is their ease of extraction from flight controllers or MAVLink messages without requiring additional hardware, enabling real-time implementation and leveraging rich and diverse sources of data. Although the performances of the proposed approaches are satisfactory, internal sensor data, such as GPS, barometric, or inertial measurements, are susceptible to environmental noise (e.g., wind, GPS drift, or pressure changes), which may lead to high rates of false positives. In addition, cumulative sensor errors during flight paths pose a challenge for reliable detection. To enhance the robustness of the proposed models and make them more efficient, training scenarios must incorporate diverse flight paths and perturbations under both normal and attack conditions. More effort should be devoted to studying the generalizability of the proposed models.

Internal circuitry properties

Circuitry data, such as time series measurements of voltage, current, and resistance, can be a valuable source of information to detect hardware-based attacks. Almeida et al. (2024) investigated the changes in impedance, reactance, and resistance in UAV control circuits to detect Hardware Trojan (HT) attacks. Physical modifications to UAV hardware, such as adding malicious circuits or altering motor controllers, often disrupt Pulse Width Modulation (PWM) signal generation or propagation. For example, a HT in a motor driver could alter duty cycles, leading to unpredictable UAV behavior that can be detected through PWM analysis, as suggested by Rahman et al. (2020). A different perspective was adopted in (Alva et al. 2024), where the authors investigated the relation between deviations in PWM signals from UAV rotors and the detection of GPS spoofing attacks.

Feeding internal circuitry features to ML models for cyberattack detection in UAVs offers key advantages, including their hardware-level generation, which makes them resistant to software-based attack obfuscation, thus enhancing resilience against sophisticated threats like adversarial ML evasion. This approach is harder for attackers to manipulate without physical access to UAVs, and disadvantages include the need for platform-specific adaptations due to the diverse designs, components, and manufacturers of UAVs, which result in varied circuitry properties. A detection model trained on one UAV's hardware signature may not generalize across different chipsets or motor controllers, requiring costly and complex customization for each configuration.

External signal characteristics

The signal properties in a UAV's GPS receiver are excellent sources of data for detecting GPS spoofing attacks as well as jamming against drones. Wei et al. (2024) proposed an approach that relies on Carrier-to-Noise Density Ratio (CN0) and Doppler frequency for attack detection. Meanwhile, Sung et al. (2022) proposed a DL model trained on Signal-to-Noise Ratio (SNR) features. The approach registered satisfactory results when the windows size, e.g. the duration over which the signal and noise power are averaged to compute SNR, is between 8 and 12 ms, enabling rapid sampling but posing challenges related to increased computational demands. Although introducing a slight computational overhead, as noted by Wei et al. (2024), numerical GPS signal features can be leveraged to develop lightweight and computationally efficient ML models for the detection of GPS spoofing.

The numerical characteristics of transmission signals as well as their spectrogram images are employed by researchers to develop ML approaches for the detection of data link jamming. In particular, Li et al. (2022a) proposed an approach leveraging key Orthogonal Frequency Division Multiplexing (OFDM) parameters, such as signal power, SNR, and energy threshold. Their findings demonstrate that using spectrogram images instead of numerical signal features can improve detection accuracy by 7.6%. However, this enhancement introduces a trade-off: numerical feature-based models are computationally lightweight, while spectrogram-based models imply higher computational costs due to image processing requirements.

OFDM features are particularly valuable to detect data link jamming in UAVs serving as aerial base stations for 5 G connectivity. These features include the Signal-to-Interference-plus-Noise Ratio (SINR) and Received Signal Strength Indicator (RSSI) (Viana et al. 2022b, 2023), as well as SNR measurements within the UAV resource block Viana et al. (2022a).

In addition to RSSI, angle of arrival (AoA) characteristics were used by Aladi and Alsusa (2023) to verify the legitimacy of the received signals, allowing detection of identity spoofing. The Received Signal Strength Difference (RSSD) and the Time Difference of Arrival (TDoA) between two radio signals were leveraged by Chulerttiyawong and Jamalipour (2023) to detect Sybil attacks in FANET-based UAS networks.

A major limitation of external signal characteristics for attack detection in UAS is their sensitivity to environmental conditions, which can lead to false positives or reduced detection accuracy. These features depend heavily on factors like weather (e.g., rain, fog), and interference from other devices (e.g., Wi-Fi or radio signals). For effective detection, this requires massive data collection under diverse environmental conditions to train robust ML models.

Network traffic characteristics

Network traffic characteristics can be indicative of attacks in UAS communication networks, particularly within the upper layers (e.g., application, transport, or network layers) of interactions between UAVs or between UAVs and GCS. These features can be derived from pcap files for the detection of DoS and Distributed DoS (DDoS) attacks (DC Bertoli et al. 2021; Sharma et al. 2024). They can also be derived from MAVlink packets to detect MITM attacks, such as in (DC Bertoli et al. 2023), or wormhole and sinkhole attacks, such as in (Renu and Sarveshwaran 2024). For example, the features considered in (DC Bertoli et al. 2021) include the type of protocol, the frame length, the time delta between the source and the destination for the detection of deauthentication, UDP and TCP flood in Wi-Fi connected UAV.

According to DC Bertoli et al. (2023), temporal-statistical features offer significant advantages for detecting anomalies in UAS networks. These features, including communication jitter or the count of packets in a defined time window, excel at identifying deviations in network behavior. For deterministic message-based protocols with predictable message timings (e.g. heartbeats), this approach proves to be reliable across various types of protocols. For example, in an MITM attack on UAV-to-GCS communication, the authors suggest that an external entity intercepting network traffic would likely cause increased jitter in the communication link. Likewise, during a DoS attack, the packet count within a specific time window would exceed the expected norms, signaling anomalous behavior. However, different UAVs or Wi-Fi networks may use different implementations or configurations of protocols, making it difficult to generalize protocol-type features across UAS.

Payload data

UAVs can be equipped with various payloads tailored to specific mission requirements in different application domains. These payloads include cameras, external sensors, communication equipment, and scientific instruments. Data derived from these payloads have been utilized by researchers to detect various attacks against UAS, particularly in the context of drone swarms.

Naït-Abdesselam et al. (2022) proposed an approach to detect the injection of false data into images collected by a drone swarm. For each image captured by a UAV, the GCS identifies its nearest neighboring UAV using the Mahalanobis distance, then compares their CNN classification results. A mismatch between these results puts the UAVs on a suspected state. However, this approach requires broadcasting CNN classification results among UAVs, which itself is vulnerable to FDI. A potential solution to the previous limitation could be found in (Titouna and Naït-Abdesselam 2021). The authors proposed a framework based on mobile agents for data collection. The agents move between UAVs, extracting data from each node to help detect and prevent FDI by insider attackers.

Payload data can be used not only to detect FDI, but also to identify GPS spoofing in UAV operations, as proposed by Xue et al. (2020); Liu et al. (2024). The authors compared real-time aerial images from the UAV camera with historical Google Earth satellite images of the corresponding GPS locations. However, Xue et al. (2020) acknowledged the limitation of the proposed approach in large landscapes where there is no diversity of visual features and its low tolerance to destructive interference in image formation. Furthermore, Liu et al. (2024) noted that their approach is susceptible to errors in visually similar areas and conditions at night, leading to an increased false negative rate.

Relying solely on payload data for UAV cyberattack detection is problematic due to its vulnerability to adversarial attacks. Adversaries can inject subtle perturbations to mislead ML models, causing misclassification or failure to detect malicious activity. Thus, greater efforts are needed to enhance the resilience of payload-based approaches in adversarial environments, such as through adversarial learning or redundancy approaches.

The use of specific feature types for attack detection in UAS has revealed several limitations. These include challenges with generalizability across diverse systems, sensitivity to environmental conditions, and susceptibility to adversarial attacks.

To address these limitations, researchers like Hassler et al. (2024) have proposed fusing cyber and physical feature data to improve attack detection in UAS. They conducted an in-depth analysis of features using SHapley Additive exPlanations (SHAP) to evaluate their contributions, demonstrating that combining cyber features (e.g., network traffic metrics like protocol type or frame length) with physical features (e.g., sensor data such as IMU readings or flight trajectory) significantly improves the performance of cyberattack detection.

Researchers have also incorporated system logs to improve attack detection in UAS, providing valuable information on their operational state. For instance, Basan et al. (2021) developed an IDS for UAV swarms that identifies anomalies and predicts normal behavior, DoS, de-authentication, and packet spoofing attacks. Their approach combined network traffic data, CPU utilization, and temperature metrics, while also analyzing the correlation between entropy in CPU usage, network traffic patterns, and attack occurrences. Similarly, Aldossary et al. (2025) leveraged system log features, including CPU usage, memory utilization, battery levels, and network traffic characteristics such as jitter, packet loss rate, and response time, in addition to flight control and sensor data. This multi-feature integration improved the detection of cyberattacks by capturing both cyber and physical indicators of malicious activity.

Preprocessing techniques

Preprocessing is a critical step in applying ML to detect cyberattacks on UAS. It involves cleaning raw data, selecting key features to ensure ML models learn attack patterns effectively, and maintaining computational efficiency.

Data cleaning

Raw UAS data often contain noise, missing values, or outliers due to sensor errors, environmental interference, or incomplete captures during attack simulation. Data cleaning improves the learning process, leading to improved performance of ML models. Gaber et al. (2021) investigated the impact of data cleaning on the detection of DoS attacks on UAS, achieving higher accuracy and faster detection time compared to DC Bertoli et al. (2021), who did not apply data cleaning.

Null values and duplicated rows can affect the ML process by introducing noise and reducing the accuracy of ML models. In the context of ADS-B message injection attacks on UAS, collisions in ADS-B messages often generate a large number of null values, which were meticulously removed from the dataset generated in (Slimane et al. 2022a). This data cleaning step ensured a more reliable dataset, enhancing the performance of the ML models considered for detecting such cyberattacks. Korium et al. (2024) discovered high number of outliers in some features of the UAV attack dataset (Whelan et al. 2020). These features are generally removed, as in (Tlili et al. 2023). Rows containing missing or null values can also be eliminated (Park et al. 2020). In contrast, Khoei et al. (2023) performed mode imputation, where missing values were replaced by the most frequent value to maintain the richness of the dataset provided in (Aissou et al. 2021). Meanwhile, Whelan et al. (2022) opted for linear interpolation to deal with NaN values.

The choice of techniques to handle outliers, missing values, null values, or NaN values in UAS cyberattack detection depends on factors such as the type of attack and the size of the dataset. For example, in detecting GPS spoofing attacks, outliers may represent attack indicators, such as extreme positional deviations, and should be retained. Furthermore, with limited data, the trade-off between data retention and model robustness must be carefully balanced to ensure effective detection.

Feature engineering

Feature engineering involves the manual or algorithmic creation and transformation of features to optimize ML model performance. This process is critical for distinguishing normal UAS behavior from malicious activities, as raw data are often unsuitable for direct use in ML models. Feature engineering includes techniques such as feature extraction and data transformation to produce robust and exploitable data representations.

Feature extraction

Feature extraction is the process of automatically deriving new features from raw data, often transforming unstructured or high-dimensional data (e.g., images, text, network pcaps) into a more compact or meaningful representation.

Convolutional Neural Networks (CNN) are among the most prevalent algorithms for extracting feature data from UAS. Wu et al. (2023) employed this DL algorithm to derive spatial status data features from UAV sensor inputs. Similarly, Sun et al. (2023) applied a 1D-CNN to identify the most relevant features from a GPS spoofing dataset, incorporating the Parametric ReLU (PReLU) activation function to improve the processing of convolution operation output. Furthermore, Ma et al. (2024) utilized a CNN with PReLU to mitigate overfitting, thereby improving the generalizability of the model. However, CNN models require significant computational power and memory, especially for processing high-dimensional data from UAS sensors, rendering their deployment on resource-constrained UAV platforms impractical. In contrast, Chen et al. (2025) performed feature extraction by applying summarization techniques to consolidate the features of ten packets into a single representative sample. This approach mitigates the risk of overfitting during the training process by decreasing sensitivity to variations in individual packets.

Data transformation

Data transformation includes various processes to convert raw data into a format suitable for modeling or analysis. This begins with encoding categorical variables using methods such as one-hot encoding or label encoding. Subsequently, the numerical data can be normalized using techniques such as standardization (Slimane et al. 2022a) or min-max scaling, ensuring compatibility with ML algorithms and improving convergence during training.

Depending on the nature of feature data and the approach adopted, significant transformations of initial datasets have been undertaken by some researchers. For example, GPS signal samples were transformed into RGB images by Korium et al. (2024) to enable detection through image processing. A more minor transformation was introduced by Naït-Abdesselam et al. (2022), in which the images collected by the UAVs were resized using Nearest-Neighbor Interpolation (NNI) to detect falsified data within the images collected by a swarm of drones. In contrast, new features, such as payload entropy and variance, were derived from existing features by (Chen et al. 2025) to help improve the performance of the detection of network attacks.

Feature selection/reduction

A diverse set of features can be extracted from UAS, with research such as (Brewington and Kar 2023) extracting up to 500 sensor attributes. Utilizing the complete set of these features for training ML models is computationally impractical due to the high dimensionality and corresponding resource demands. Feature selection is therefore critical, as it enables the identification of the most discriminative features, thereby reducing computational complexity and processing time, while enhancing model generalization capability.

For example, Wu et al. (2023) determined the minimal features required to optimize computational efficiency while maintaining high detection accuracy. Specifically, they identified nine features for DoS detection, achieving a detection rate of 99.4% and six features for GPS spoofing detection, yielding a detection rate of 99.1%. Furthermore, Park et al. (2020) removed control-specific features to improve model generalizability, inferring that their approach is applicable across diverse UAV platforms without compromising performance.

In this part, we discuss some feature selection and feature reduction techniques used by researchers within the preprocessing phase.

-Principal Component Analysis (PCA): Certain ML models, particularly unsupervised algorithms, including One-Class Support Vector Machine (OC-SVM), Isolation Forest (IF), and Local Outlier Factor (LOF), exhibit degraded performance on high-dimensional datasets, as noted by Fraser et al. (2021). To address these challenges, dimensionality reduction is essential for high-dimensional datasets. PCA has been widely employed by scholars to transform datasets into a lower-dimensional space, maximizing variance, and reducing redundancy. Examples of studies include (Khalil and Rahman 2023; Korium et al. 2024; Brewington and Kar 2023; Badar et al. 2025). However, a major limitation of PCA is the loss of interpretability and potential inclusion of noise in Principal Components, which can limit its utility in some UAS-specific applications.

-Filtering by feature importance: The presence of irrelevant features in a dataset can significantly degrade the classification performance, which requires robust filtering methods. This filtering process begins with a feature importance analysis, which ranks features according to their predictive utility. Subsequently, only a subset of the most significant features, typically a predefined percentage, is retained for model training. Hakeem et al. (2025) employed the Chi-square feature selection technique to identify and retain the most relevant features, substantially improving the efficiency and accuracy of the detection process. Their approach resulted in a marked improvement in the performance of the MobileNet model, with the classification accuracy increasing from 90.78% to 98.49%. Wei et al. (2025) employed a Random Forest (RF) model to evaluate feature importance, quantifying the decrease in node impurity (using the Gini index) during decision tree splits and ranking features based on their contribution across the ensemble. Similarly, Mouatassim et al. (2025) applied mutual information to quantify the dependency between features and the target variable, effectively identifying features with high predictive power. Furthermore, Slimane et al. (2022b) used the Extra Trees classifier, which computes feature importance through randomized tree construction, to select features for GPS jamming detection in UAS.

-Recursive Feature Elimination (RFE): This method is an iterative feature selection technique that systematically identifies and retains the most relevant features from the training dataset. For example, Ihekoronye et al. (2024) employed RFE with a Decision Tree (DT) model as an estimator to rank and select features. Similarly, Al-Syouf et al. (2025) utilized RFE with a Gradient Boosting Machine (GBM) as the estimator. The effectiveness of RFE depends on the choice of estimator and may be unsuitable for on-board UAV deployment due to its high computational cost.

- Correlation analysis: Highly correlated features in the dataset can affect the performance of the classification models. Accordingly, correlation analysis is a key feature selection technique that evaluates inter-feature relationships in a dataset using statistical measures such as Pearson, Spearman, or Kendall correlation coefficients. Spearman rank (Schober et al. 2018) and Pearson correlation analysis (Benesty et al. 2009) are the most widely used techniques. For example, Wei et al. (2025); Aissou et al. (2021); Sun et al. (2023) used Spearman correlation. In contrast, Gasimova et al. (2022); Da Silva et al. (2023); Hong and Yoo (2024) employed the Pearson correlation. Meanwhile, the Kendall correlation technique was employed in (Slimane et al. 2022a). To gain comprehensive and distinct insights into the relationships between features and the target variable, Badar et al. (2025) assessed feature relevance using chi-square tests, Pearson correlation coefficient, and ANOVA (Analysis of Variance).

Unlike Pearson's correlation, which assumes normality and relies on raw feature values, Spearman and Kendall coefficients use rank-based methods, making them robust to non-normal distributions; however, all correlation methods require sufficient data volume to ensure reliable estimates.

- **Genetic Algorithm (GA):** Selecting the appropriate features can also be achieved using evolutionary algorithms, such as the GA adopted by Mehmood et al. (2022), or the Greedy-based Genetic (GG) algorithm used by Miao et al. (2024).

Both GA and GG algorithms are effective feature selection techniques for attack detection in UAVs, as they can optimize feature subsets for improved detection accuracy and efficiency. While full GAs are better suited for offline processing due to their computational complexity, GG algorithms offer a more practical option for on-board or hybrid implementations. By leveraging offline computation, hardware optimization, and lightweight models, these algorithms can be successfully applied to enhance attack detection on resource-constrained UAVs.

- **Fuzzy Rough Set (FRS)**: FRS has been employed for feature selection due to its robustness and effectiveness in managing incomplete and imprecise datasets, and its algorithm performs detection through mathematical computations, which provides a high level of interpretability. Wu et al. (2024) proposed an improved FRS model to reduce features. The authors highlighted the high performance of the proposed approach by emphasizing the reduction in runtime achieved through proper feature selection.

Dataset balancing

Datasets specific to UAVs are often imbalanced, with the benign class being predominant. Consequently, imbalanced datasets are solved mainly by oversampling the minority class or undersampling the majority class, such as randomly choosing samples as in (Da Silva et al. 2022).

Several dataset balancing techniques have been leveraged by researchers. Wu et al. (2023) adopted a sampling-interpolation process. The technique involves undersampling benign data and interpolating attacked data by averaging adjacent pairs to create new synthetic samples inserted between them, balancing the dataset simply but potentially losing information from undersampling and introducing noise through basic interpolation.

Furthermore, Synthetic Minority Oversampling Technique (SMOTE) (Chawla et al. 2002) has been widely adopted in surveyed studies to optimize model performance (Ihekoronye et al. 2024; Badar et al. 2025; Ma et al. 2024). The technique generates synthetic attacked samples by interpolating between minority class instances and their k-nearest neighbors (KNN), preserving more data nuance than sampling-interpolation but risking overfitting to noisy neighbors. SVM-SMOTE was used by Sun et al. (2023) to improve generalizability. The technique extends SMOTE by focusing synthetic sample generation near the SVM decision boundary, improving classifier robustness for GPS spoofing detection but potentially increasing computational cost compared to basic SMOTE for high-dimensional datasets.

However, these data augmentation methods are often associated with overfitting, where the model is fitted to the noise in the data (Huang et al. 2022). Thus, SMOTE has been combined with Edited Nearest Neighbors (ENN) to reduce noise by removing samples whose class labels differ from the majority of their nearest neighbors. However, the application of ENN may inadvertently eliminate informative samples, potentially reducing the diversity of the dataset, as noted by Korium et al. (2024). Consequently, the latter researchers preferred to use the Adaptive Synthetic Sampling algorithm (ADASYN) (He et al. 2008) to oversample the minority class. Other relevant techniques include Gradient-Based Adaptive Resampling (GAR) used by Aldossary et al. (2025) and data augmentation through Generative Adversarial Networks (GANs) (Huang et al. 2022).

The choice of data augmentation techniques depends on the characteristics of the dataset, such as noise level and feature heterogeneity. In general, much effort is needed to find suitable data augmentation techniques that take into account the specificity and diversity of feature data in UAS.

Furthermore, since UAVs often operate in complex and unpredictable environmental conditions, such as wind fluctuations, electromagnetic interference, and temperature variations, as noted by Wei et al. (2025), preprocessing techniques are critical to improve model reliability. Although these techniques can enhance model generalization capability, improve performance, reduce computational costs to some extent, and prevent overfitting to minor classes, they may result in a loss of interpretability or introduce additional noise, potentially limiting their utility in a UAS context. Moreover, given that UAVs have limited computing resources, implementing preprocessing techniques for online training on-board these vehicles requires optimizing ML algorithms to be lightweight and computationally efficient. More research efforts are needed to optimize preprocessing techniques tailored to UAS.

Learning paradigms

Learning paradigms are fundamental to ML approaches, shaping how models are designed and trained to detect attacks. Their importance lies in guiding the selection of appropriate techniques and strategies to address the complex and dynamic nature of cyber threats in UAS.

In this subsection, we examine and compare the learning paradigms used by researchers to detect attacks on UAS. We categorize these paradigms into supervised learning, unsupervised learning, and Reinforcement Learning (RL). Additionally, we treat adversarial learning as a distinct paradigm because of its unique mechanisms and objectives.

Supervised learning

Supervised learning is a reliable approach for detecting attacks in UAS, leveraging labeled datasets to train models that can accurately identify malicious activities such as GPS spoofing, jamming, or UAS network attacks. These models, including traditional ML, DL, and ensemble learning techniques, have been extensively explored in the literature for their ability to enhance UAS cyberattack detection.

Traditional Machine Learning

Traditional ML models, including DT, Logistic Regression (LR), Naive Bayes (NB), and SVM, were selected by researchers for their efficiency, interpretability, and robust performance in cyberattack detection. These models are particularly well-suited for UAS, where computational resources are typically limited.

DTs are highly interpretable, lightweight, and fast predicting models. They were proposed in Ihekoronye et al. (2024); Moustafa and Jolfaei (2020); Nayfeh et al. (2023). Compared to DT, SVM excels in classifying high-dimensional data but requires careful kernel and parameter tuning, making them less suitable for complex temporal patterns in UAV data and more computationally intensive. Bayrak (2024) proposed a linear SVM model to detect anomalies in UAV communication traffic. Meanwhile, Slimane et al. (2022a) proposed an ADS-B message injection detection technique based on a C-SVM model. Furthermore, Aissou et al. (2022b) proposed nu-SVM for the detection of GPS spoofing attacks. Although the recorded performance seems to be satisfactory, the probability of misdetection is relatively high.

The previous approaches were employed for binary or multiclass classification. In contrast, Alkhatib et al. (2024) proposed a multi-output multiclass to detect and locate GPS jamming attacks. The authors trained simple multilayer perceptron (MLP), RF, KNN, LR, DT, NB and SVM on a custom-made dataset. The proposed models can classify the GPS signal into five classes, namely normal, barrage, single-tone, succession pulse, and protocol-aware jamming (Lichtman et al. 2016), in addition to indicating the direction and distance separating the UAV and the attacker. The authors concluded that MLP outperformed other ML algorithms, while DT produced the lowest prediction time. However, the generalization capability of MLP is superior to DT, as highlighted by (Greco et al. 2021).

Deep learning

DL approaches have been increasingly applied to detect attacks in UAS, leveraging their ability to model complex patterns in data. In this part, relevant studies are organized on the basis of their underlying architectures, namely core and hybrid architectures.

- Core architectures: Core DL architectures employed by researchers include CNN, Long Short-Term Memory (LSTM), Transformers, and Graph Neural Networks (GNN), among others.

Designed for spatial data, CNN effectively extracts spatial features such as patterns and anomalies in UAS sensor inputs and leverages its convolutional layers to distinguish critical attack-related characteristics from normal UAS operational data. Researchers have explored various CNN architectures to detect attacks in UAS. Sung et al. (2022) developed a 1D CNN based on the ResNet architecture, leveraging its deep structure and residual connections to capture complex patterns in GPS signal data. In contrast, Abu Al-Haija and Al Badawi (2022) adopted a shallow CNN architecture, prioritizing simplicity and reduced computational complexity. Meanwhile, Hadi et al. (2024) employed a feedforward CNN architecture, combining ReLU and Tanh activation functions to balance non-linearity and gradient stability for effective attack detection. Furthermore, Li et al. (2022a) proposed using EfficientNet-B0, a lightweight CNN architecture optimized for efficiency, to train on spectrogram images for detecting GPS spoofing attacks in UAS, achieving high performance with minimal computational resources.

In contrast to CNN, which processes input through layered feature extraction, capsule networks leverage dynamic routing to capture spatial hierarchies, enhancing performance in complex tasks. Talaei Khoei et al. (2025) demonstrated that the capsule family outperforms other techniques in binary and multiclass problems to detect GPS spoofing attacks. Among these, Efficient-CapsNet achieved the best results for both binary and multiclass tasks. In contrast, CNN delivered the poorest performance compared to Efficient-CapsNet, CapsNet, and Da-CapsNet in these scenarios. However, capsule-based models typically require higher computational resources and longer inference times due to their complex routing mechanisms.

Furthermore, LSTM networks have been utilized to address temporal dependencies in UAS attack detection. For example, to detect interception of communication exchanges between a UAV and a GCS through MITM attacks, Zhang et al. (2022) proposed a framework combining Wavelet Leader Multifractal Analysis (WLM) with an LSTM architecture. This framework includes a 1D sequence input layer, a Bidirectional LSTM (BiLSTM) layer with 500 hidden units, a fully connected layer, and a softmax layer. LSTM effectively captures temporal dependencies in communication sequences, leveraging WLM to enhance feature extraction of multifractal patterns indicative of MITM attacks, thus distinguishing malicious interceptions from normal UAS communications. Optimized for time series data, LSTM was used by Tlili et al. (2023) to detect intrusions in UAVCAN message exchanges within UAVs, by Ding et al. (2021) to detect sensor-based and flight control attacks, and by Sun et al. (2023) to detect GPS spoofing attacks.

In contrast to LSTMs, which process input sequentially, transformers leverage parallel processing, resulting in reduced training time. Sharma et al. (2024) used a Time Series Transformer (TST) to detect DDoS attacks in UAS. The TST outperformed several ML and DL models, including XGBoost, IF, LSTM, BiLSTM, and LSTM-Attention. However, transformer-based models are resource-intensive, with TST exhibiting the highest memory usage among the compared models and a longer inference time because of its multiple encoder layers.

Unlike CNN and LSTM which provide direct classification, Siamese networks focus on similarity learning and are ideal for tasks with limited data per class. As an example, Xue et al. (2020) proposed Siamese SqueezeNet to compare historical satellite images of the UAV's GPS location with real-time aerial images taken by its camera for the detection of GPS spoofing attacks.

GNN-based methods outperform traditional models by capturing topological relationships and local features in UAV swarm communication, enabling accurate anomaly detection. The GNN-based IDS proposed by Mughal et al. (2024a) modeled UAVs as graph nodes with spatial (position, orientation, distances) and temporal (velocity, acceleration, control signals) features to detect cyberattacks such as FDI, evil twin, replay, and DoS. Compared to LSTM and 1D-CNN, the GNN showed superior performance due to its ability to model complex inter-UAV relationships and dynamic patterns, which other models cannot effectively handle.

- **Hybrid architectures:** These architectures are designed to overcome limitations of core architectures. They are more resource-intensive, but often yield better performance.

Researchers used DNN as the core architecture to detect intrusions in UAS. Hadi et al. (2024) proposed a multi-tier DNN fusion model, integrating MLP, CNNs, and GANs as core components. In this hierarchical framework, the most accurate DNN classifier is positioned first to categorize samples into well-classified and misclassified groups. The misclassified samples are then passed to the next most accurate DNN classifier for further learning and classification, with subsequent classifiers following the same process. This tiered approach improves classification accuracy compared to the standalone MLP, CNN, and GAN models.

Furthermore, some scholars used hybrid approaches that integrate Graph Convolutional Network (GCN) with other core architectures. To identify GPS spoofing attacks on a UAV swarm, Dang et al. (2023) employed GNN, made up of GCN layers combined with recurrent network layers, analyzing the discrepancies between the GPS topology information of the swarm and its communication topology. Another study was proposed by Du et al. (2025) to detect intrusion in UAV controller networks. The approach integrates LSTM and GCN, leveraging the efficiency of GCN in capturing network topology and the capabilities of LSTM in processing time-series data. However, the model's ability to detect replay attacks was relatively weaker due to high data similarity.

Attention mechanisms in hybrid architectures improve performance by focusing on relevant features. For example, Wu et al. (2023) proposed an interpretable framework in which features are processed by a BiLSTM network to address long-term dependencies. These features are then passed on to an attention mechanism that focuses on outputs most relevant to the ground truth for detection tasks.

Similarly, the Adaptive Neuro-Fuzzy Inference System (ANFIS) proposed by Khalil and Rahman (2023) takes advantage of its mathematical foundation to improve decision-making for attack detection. The researchers employed a Takagi-Sugeno-Kang (TSK) fuzzy inference system to establish the IF-THEN rules that are combined with an Artificial Neural Network (ANN) to make decisions. This framework enhances interpretability, but exhibits limited generalizability.

Other hybrid approaches used by researchers involved the use of CNN-LSTM architectures, such as in the (Viana et al. 2022a) study that used three convolutional layers, one LSTM, one dropout, a fully connected layer, and an output layer for classification of jamming attacks. As CNN and LSTM have more significant temporal complexities, particularly for larger datasets, suggesting their inability to handle massive data, some authors proposed hybrid architectural enhancements. Aldossary et al. (2025) proposed a Cross-Layer Convolutional Attention Network (CLCAN) adaptive architecture that uses multi-scale attention mechanisms and dynamic feature fusion to reduce processing demands.

Ensemble learning
Ensemble learning models are highly effective for detecting attacks in UAS. By combining multiple ML models, such as DT, RF, or Gradient Boosted Decision Trees (GBDT), these ensembles can identify patterns of malicious behavior, such as signal jamming, GPS spoofing, or DoS attacks, with greater accuracy than individual models.

RF generally offer strong generalization capabilities and interpretability, but their ability to handle long-term temporal patterns may be outperformed by deep architectures designed explicitly for sequence modeling. Baig et al. (2022) introduced an RF-based method to detect hijacking, GPS signal jamming, and DoS attacks targeting drones. Furthermore, Ferrão et al. (2024) conducted a systematic comparison of eight supervised models with regard to their application in the detection of attacks and faults in the physical components of UAVs. The authors identified RF as the best model, balancing performance and computational efficiency.

Bootstrap Aggregating (bagging) builds multiple DTs on random data subsets to reduce variance and get an aggregated predictor, stacking combines predictions from diverse models using a meta-learner for improved accuracy, and boosting iteratively trains DTs to focus on misclassified samples, enhancing overall performance but increasing computational complexity. Gasimova et al. (2022) compared these approaches, using GBDT for boosting, bagging with DTs, and stacking with KNN, NB, DT, RF, and LR as base models feeding into a meta-model. Their findings showed that stacking achieved the highest accuracy but had the slowest average prediction time per instance and the highest memory usage. A similar comparison was proposed in (Al-Syouf et al. 2025).

The researchers demonstrated that a stacking ensemble method achieves significantly higher detection accuracy than individual models when handling complex data of high dimensions (Ma et al. 2024). The authors of this study employed a stacked ensemble to classify GPS spoofing attacks in UAS. They used XGBoost as the base classifier. LR served as the meta-learner, combining the output of the XGBoost base learners to enhance the generalization and robustness of the model against complex attack patterns.

Unlike faster bagging or stacking approaches, boosting improves the accuracy of attack detection in UAS, but its slow training speed makes it unsuitable for online learning, where the model continuously updates and adapts their attack detection mechanism in real time or near real time as new data is received. For example, Tufekci et al. (2024) proposed GBDT, with a training time of 4.21 s, to detect various attack scenarios, including MITM, TCP SYN flooding, UDP flooding, and deauthentication.

Compared to GBDT which are computationally intensive, LightGBM achieves faster training and comparable or better accuracy for attack detection in UAS. Da Silva et al. (2023) proposed this algorithm to detect network attacks. On the other hand, Slimane et al. (2022b) proposed a LightGBM model to detect GPS jamming attacks. The model was faster and occupied less memory with 0.041 s and 0.016 MiB, respectively.

Some hybrid approaches have been proposed by researchers leveraging DL as well as traditional ML for ensemble learning. She et al. (2025) proposed an improved AdaBoost-CNN scheme that transfers updated network parameters from each CNN base classifier to the next during training, ensuring continuous parameter refinement while addressing the challenge of limited GPS signal samples received by UAVs. Furthermore, Wu et al. (2024) proposed a tiny ML-based IDS. To improve interpretability and reduce resources, the researchers adopted a combination of a 20 trees-RF model and a Shallow DL (SDL) model that contains only two hidden layers for multiclass traffic classification.

Unsupervised learning
Unsupervised learning is specifically evaluated for their ability to detect zero-day attacks, such as K-means, which have been evaluated in (Sajid et al. 2023). However, there has been an increasing interest in the literature devoted to unsupervised novelty detection.

Novelty detection, also known as anomaly or outlier detection, is a ML approach that identifies unusual or rare patterns in the data. It relies on the assumption that most data are "normal", and novelties deviate significantly from this norm. To detect attacks, the novelty detection approach trains algorithms exclusively on samples of normal behavior and subsequently evaluates the model on both normal and anomalous samples. Several researchers leveraged Autoencoder (AE) for attack detection in UAS. This algorithm compresses input data into a latent space (encoder) and reconstructs it (decoder). Anomalies are detected when reconstruction errors are high, as they deviate from normal patterns learned during training.

In the work of (Whelan et al. 2022), a straightforward AE was employed to effectively identify GPS spoofing and jamming attacks. For comparison, Da Silva et al.

(2023) adopted a more complex stacked AE to detect GPS-related attacks. In a different approach, Fraser et al. (2021) used a Convolutional AE (CAE), specifically tailored to detect GPS spoofing attacks with improved precision. Meanwhile, Park et al. (2020) introduced a linear AE, incorporating linear neurons and ReLU activation functions, to address DoS attacks and GPS spoofing with notable efficiency. Similarly, Aladi and Alsusa (2023) proposed an AE-based method to verify the authenticity of received signals, effectively detecting UAV identity spoofing. The researchers utilized RSSI to approximate the UAV's position and juxtaposed it with the actual distance obtained from the on-board GPS. Subsequently, they determined the location error, which serves as an input parameter for the AE model. These approaches are generally compared with the OC-SVM and LOF models to prove their superiority.

In contrast, LOF was the best-performing model, achieving an accuracy of 75.87% among the models proposed by Alva et al. (2024). The researchers introduced an IDS that detects GPS based attacks by learning from novelties in PWM signals which powers the UAV's motors. The authors devised this approach after observing the direct impact of GPS-based attacks on UAV movements, suggesting potential alterations in PWM signals.

Compared to supervised models, unsupervised models generally exhibit moderate performance, typically achieving accuracy below 93%. The authors reporting high performance emphasize the need for real UAS validation to confirm both reliability and computational efficiency. Due to their time complexity, unsupervised methods such as IF and LOF are well suited for medium-scale datasets, as noted by Chen et al. (2025). Although these models show strong potential for generalizability across various UAS hardware platforms, a key limitation of novelty detection lies in their difficulty distinguishing between attacks and UAV operational anomalies.

Reinforcement learning

RL is a robust method for detecting attacks on UAS. It enables agents to learn optimal detection strategies through trial-and-error interactions in dynamic environments, adapting to evolving attack patterns.

Researchers have used Q-learning to develop robust IDS, particularly for UAV networks. Wu et al. (2023) introduced a Q-learning-based cooperative intrusion detection system (Q-TCID) with dynamic voting and intelligent auditing, optimizing cooperative detection and energy efficiency through distinct Markov Decision Processes (MDPs), though scalability and real-time processing are limited by computational complexity. The Deep Q-Network (DQN) approach proposed by Tocci et al. (2023) for UAV IDS prioritizes rare attack detection with higher rewards and variable penalties for false positives / negatives, improving robustness, but facing sample inefficiency and vulnerability to adversarial attacks. In contrast, Islam et al. (2025) proposed Q-Network for UAV CAN networks. The approach leverages experience replay, epsilon-greedy exploration, and a target network for stable learning, but high computational costs and real-time delays hinder practical deployment. In addition, Bouhamed et al. (2021) proposed a lightweight DQL-based Intrusion Detection and Prevention System (IDPS) that uses a custom reward function to favor minority class detection, as in (Tocci et al. 2023). The approach outperformed traditional ML models with offline training, but its pseudo-environment training risks poor real-world generalization. These Q-learning approaches excel in precise and adaptive detection, but struggle with computational demands and real-time constraints.

Although Q-learning approaches dominate the reviewed studies in this category, advanced Deep RL (DRL) methods offer complementary insights. Tao et al. (2021) and Ghelani et al. (2024) employed policy gradient methods to address complex intrusion detection tasks in UAV networks, particularly for jamming attacks. Deep Deterministic Policy Gradient (DDPG) framework, proposed by Tao et al. (2021), models intrusion detection as an MDP, tested in a simulated 3D scenario with multiple attackers, offering flexibility in continuous action spaces but requiring significant training data and computational resources. Meanwhile, the RL-based Gradient Monitoring technique (RLGM) proposed by Ghelani et al. (2024) introduces dynamic weight adjustments via localized training, outperforming federated RL and DQL in accuracy, energy usage, latency, and throughput, although its focus on jammer detection limits generalizability. Both approaches handle complex, continuous tasks effectively, but their high computational demands and sample inefficiency pose challenges for real-time UAV applications, particularly in resource-constrained settings.

Furthermore, the study proposed by Yu et al. (2025) stands out for its Multi-Task MDP (MT-MDP) framework, which models attack detection and action prediction simultaneously, leveraging historical and current states to predict communication information. This approach offers efficiency in handling multiple tasks and context-aware decision-making, but its computational complexity and extensive training requirements limit real-world validation. Unlike the Q-learning and policy gradient studies, MT-MDP addresses broader, multi-objective scenarios, making it versatile but challenging to implement in resource-limited UAV networks. Although this approach pushes the limits of the applicability of RL to complex intrusion detection in UAS, it shares the common limitation of high computational overhead, suggesting a need for hybrid methods combining lightweight Q-learning with multi-task flexibility to improve scalability and real-time performance.

Incorporating advanced mechanisms like multi-head attention and multi-task reinforcement learning increases the computational complexity. This might pose challenges for real-time implementation, especially on resource-constrained UAV platforms.

Adversarial learning

Adversarial learning has emerged as a powerful approach to counter attacks against intrusion detection mechanisms in UAS. Researchers have employed GAN to train models on adversarial samples, allowing models to resist perturbations and significantly improving their robustness against sophisticated adversarial attacks (Kostage et al. 2025).

Researchers use GAN to build generative datasets, particularly for the detection of GPS spoofing attacks. For example, El alami and Rawat (2024) generated two synthetic datasets using GAN and transformers. In fact, the generator in a GAN utilizes random noise to create realistic threat or malfunction scenarios. Meanwhile, the discriminator employs the MultiHeadAttention mechanism to systematically differentiate between genuine and synthetic sequences, improving its accuracy during training. The authors reported that the model performance was not affected by Gaussian noise. Similarly, in Alhoraibi et al. (2024), a GAN model was used to mimic spoofing signals and was tested for its ability to generate various adversarial samples, which were evaluated against the CNN and LSTM model for attack detection. Training with adversarial samples resulted in an increase of 12% in the average F1 score. Furthermore, Brewington and Kar (2023) proposed Bi-GAN, which is an extension of the vanilla GAN framework, enabling bidirectional learning and offering advanced generative modeling.

In general, supervised learning outperforms unsupervised learning, as noted in Khoei et al. (2022), but its reliance on annotated training sets limits its effectiveness in dynamic or evolving operational contexts where sensor-based attack techniques may differ from those in the original datasets. In contrast, novelty detection through unsupervised learning enables greater generalizability across UAV platforms but may misclassify anomalies within UAS as attacks. Meanwhile, RL enables adaptive learning of decision policies through interaction with the environment, offering greater flexibility in handling evolving attack scenarios and changing conditions compared to static pre-trained models. However, RL requires significant training time and robust simulation environments to be effective. Additionally, adversarial learning can enhance model resilience against attacks on security measures, but the computational complexity of GANs can pose challenges for practical deployment.

Overall, DL models surpass traditional ML models within learning paradigms, as reported by Sun et al. (2023), but their memory-intensive nature presents challenges. Techniques such as pruning and weight quantization can effectively reduce parameter counts while preserving predictive performance (Al-Sabbagh et al. 2025). However, the quantization process can introduce a loss of precision, affecting the accuracy of the model, as noted by Wei et al. (2025). Hybrid approaches, which integrate lightweight DL and traditional ML, balance performance and efficiency (Badar et al. 2025).

Furthermore, training ML models once is typically insufficient due to the dynamic nature of certain attacks, such as the non-stationary characteristics of GPS signal distributions over time, necessitating periodic retraining (Alhoraibi et al. 2024). Some researchers argue that periodic offline learning outperforms online learning for power optimization (Bouhamed et al. 2021). Others propose transfer learning to leverage prior knowledge from a source task for few-shot tasks (Hakeem et al. 2025), and to reduce computational overhead (Korium et al. 2024).

Examining ML approaches used by researchers to address the research problem reveals certain trends and statistics. Supervised learning is the dominant ML paradigm, adopted in 81% of studies addressing UAS attack detection, as shown in Fig. 8. Within this category, DL algorithms are the most prevalent, used by 46% of researchers, followed by traditional ML methods (29%) and ensemble learning paradigms (26%), basically tree-based algorithms such as XGBoost, LightGBM, and GBDT. In contrast, only 8% of studies employ RL, and just 3% explore adversarial learning, reflecting their limited adoption in this domain.

Regarding the ML algorithms which have been presented in the reviewed studies, CNN, LSTM, DT, XGBoost and AE are the most commonly adopted for detecting cyberattacks on drone systems, as depicted in Fig. 9.

Moreover, the literature does not consistently address all types of attacks. As shown in Fig. 10, GPS spoofing is the most extensively studied attack, with 38 papers focusing on its detection. However, certain attacks, such as identity spoofing, wormhole, ADS-B, and Sybil attacks, are rarely explored. Additionally, while attacks like command injection, buffer overflow, unauthorized root access, keylogger attacks, and malware infections have been exploited in the UAS context, as shown in Table 2, there is a notable lack of ML-based detection solutions for these attacks in the literature.

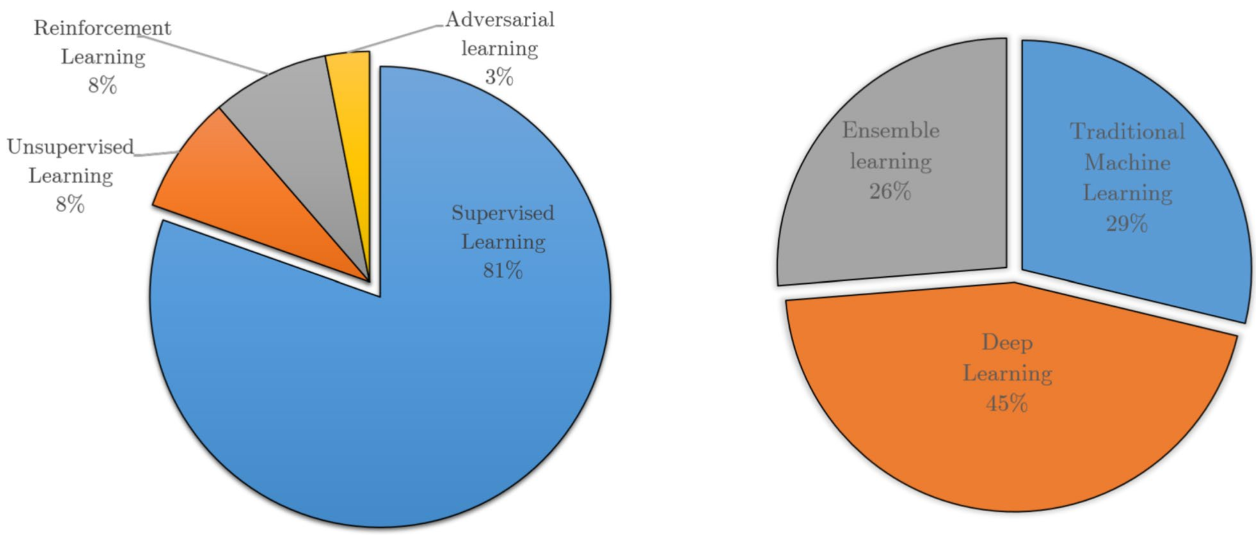

Fig. 8 Distribution of the reviewed studies: Overview of ML techniques

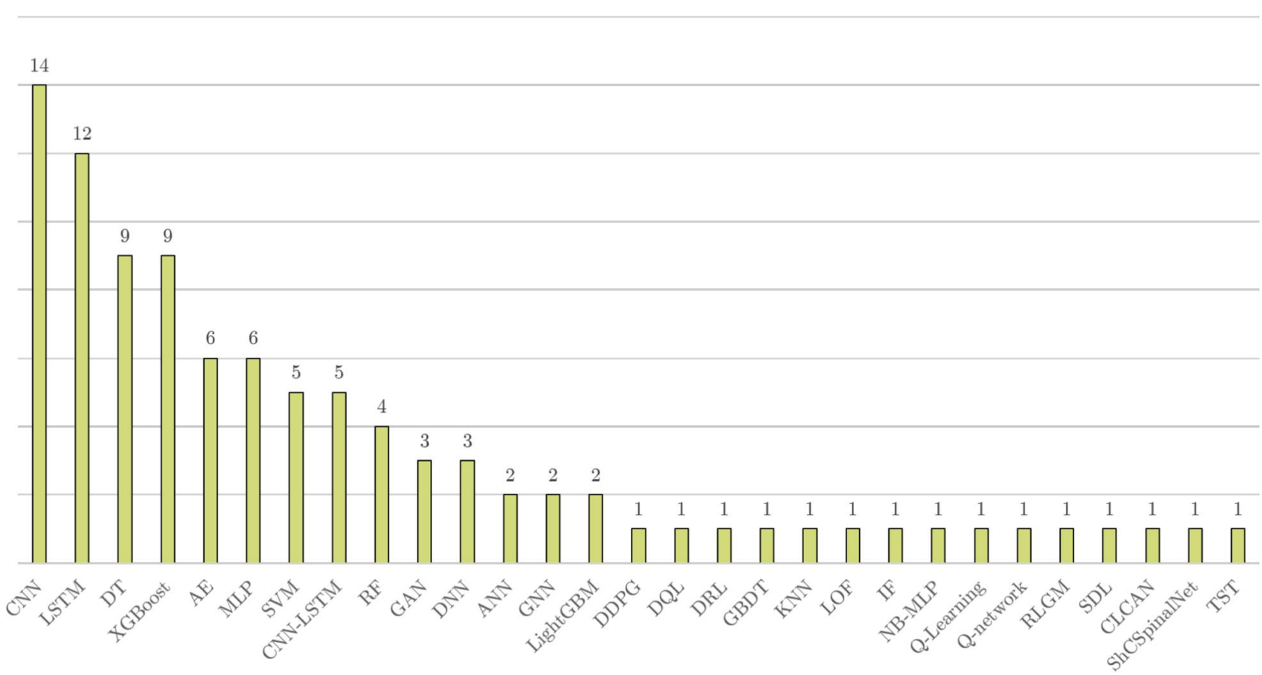

Fig. 9 Distribution of the reviewed studies: Overview of adopted ML algorithms

Evaluation

Intrusion detection can be framed as a classification task, where the performance of classifiers is assessed by comparing actual and predicted labels using a confusion matrix (Table 5). In this context:

- **True Positive (TP)** refers to an attack sample correctly identified as an attack;
- **False Negative (FN)** occurs when an attack sample is misclassified as normal;
- **True Negative (TN)** denotes a normal sample correctly identified as normal;
- **False Positive (FP)** represents a normal sample incorrectly classified as an attack.

The following evaluation metrics are generally adopted to evaluate ML models:

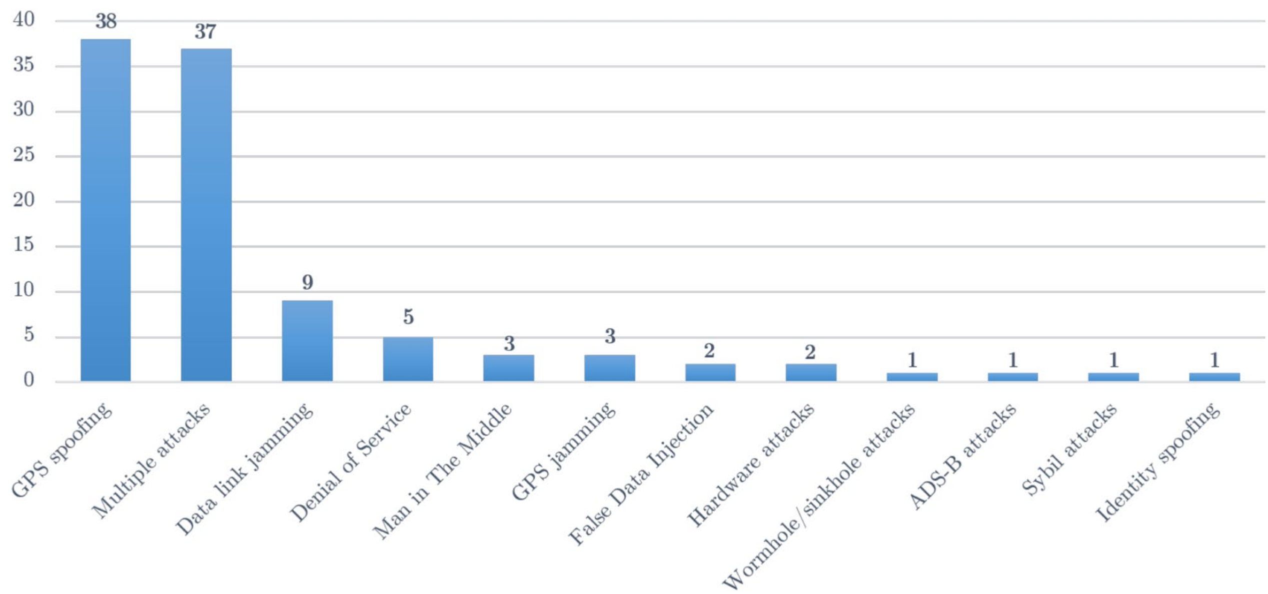

Fig. 10 Distribution of reviewed studies per type of attacks considered

Table 5 Confusion matrix for intrusion detection

		Predicted class	
		Normal	**Attack**
Actual Class	Normal	TN	FP
	Attack	FN	TP

- Accuracy is the number of correctly predicted samples divided by the total number of predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

- Precision is the fraction of correctly identified attacks out of all alerts triggered by the system :

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

- Recall is the fraction of actual attacks correctly identified by the system out of all existing attacks:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

- The F1 score is the harmonic mean of precision and recall, balancing the trade-off between the two :

$$F1\_score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{4}$$

- The Matthews Correlation Coefficient (MCC) is a robust metric for evaluating binary classification performance. It is particularly valuable in the context of attack detection in UAS, where datasets are often imbalanced. Unlike accuracy, which can be misleading in such scenarios, MCC takes into account all four components of the confusion matrix:

$$\mathrm{MCC} = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{5}$$

To assess the impact of ML-based attack detection on UAS performance, computational overhead is evaluated by comparing resource usage with and without the ML model deployed. The following metrics are typically used:

- Memory overhead quantifies the additional memory required by the system when the intrusion detection model is active, compared to a baseline configuration without the model.

$$\text{Memory Overhead (\%)} = \frac{M_{\text{with model}} - M_{\text{baseline}}}{M_{\text{baseline}}} \times 100 \tag{6}$$

- Energy overhead is the additional energy consumed due to the operation of the model during inference :

Table 6 Primary studies

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Karimibiuki et al. (2019)	Predicting abnormal behavior to authenticate autonomous UAVs though their attitude including yaw, roll and pitch	Custom dataset	OC-SVM, KNN, LR	KNN: Accuracy 93.4%, detection time 3 s	- False positive rate would be high when implemented - Limited to UAS with predefined paths
Xue et al. (2020)	DeepSim: comparing historical GPS related satellite images to UAV aerial images taken in real time to detect GPS spoofing	SatUAV dataset	Distance Threshold & Siamese Nets	Siamese SqueezeNet on RPi: Accuracy 82.6%, F1 83.7%, power consumption 1.1 W	+ Implementation on RPi + No additional hardware - Detection limited in large views - Low tolerance to destructive interference
Wang et al. (2020)	UAV path prediction by IMU information, then path compared to received GPS positions to spot deviations from planned path	Simulated data	LSTM	LSTM: detection ratio 78%, detection time 3 s	+ MATLAB simulation for evaluation - Detection limited to UAV with predefined paths - Unique metric: detection ratio
Feng et al. (2020)	XGBoost trained off-board using GPS & IMU data, fine tuned by GA, then uploaded to be trained again on-board the UAV to detect GPS spoofing	Custom dataset	GA-XGBoost, SVM	GA-XGBoost: correctness 100% in normal flight & 96.3% in attack cases	+ Validation experiments conducted on a real UAV + Adaptability to different UAV platforms - Unique metric: prediction correctness
Park et al. (2020)	Detecting DoS attacks as well as GPS spoofing attacks through novelty detection	UAV ATTACK	Linear AE	NA	+ Generalizability over other platforms - No quantitative metric
Moustafa and Jolfaei (2020)	Autonomous IDS designed to identify network attacks targeting UAS	Custom dataset	DT, SVM, NB, KNN, MLP	DT: Accuracy 99%, F1 99%, detection time 0.028 s	+ Well-detailed testbed - Traffic related to UAVs was not provided significant attention
Rahman et al. (2020)	Detection based on PWM signals and on-board sensor signals	NA	NA	NA	+ Directions on dataset collection & IDS deployment - Study limited to a preliminary discussion of the proposed approach
Bouhamed et al. (2021)	Detection & prevention of intrusions in a fleet of UAVs based on periodic offline training	CIC-IDS2017	DQL, LR, ANN, RF, NB	DQL: Accuracy 99.73%	+ Gain of 2 min in flight autonomy per charge cycle compared to online learning - No real implementation of the solution
Basan et al. (2021)	Detecting & classifying traffic into normal, DoS, deauthentication and packet spoofing	Custom dataset	MLP	NA	+ Discussion of the correlation between entropy in CPU utilization, network traffic, and the existence of attacks - No precise results
Ding et al. (2021)	Detection of sensor-based attacks and attacks targeting flight control that lead to unsafe status	Custom dataset	LSTM	Average runtime overhead 2.6% & memory size overhead 0.34%	+ Implementation on a real UAV + Computational overhead evaluation - No evaluation of the model's accuracy & F1 - Unique model

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Agyapong et al. (2021)	Detection based on telemetry data from IMU and other sensors using binary and one-class DL classifiers	Custom dataset	LSTM, LSTM-AE	LSTM: Accuracy 97.79%, detection time less than 4 ms	+ Comparing UAV-generalized to UAV-specific detectors + Implementation on RPi and a Neural Computing Stick - Spoofing activated after three minutes from take-off
Fraser et al. (2021)	Digital Twin architecture to model the normal behavior of UAVs and detect anomalies	UAV ATTACK	OC-SVM, IF, LOF, CAE, LSTM AE	CAE: F1 98.51%, AUC 100%	+ Novel approach - Higher performance - No real hardware implementation to verify computational fidelity
Aissou et al. (2021)	Comparison of Tree-based ML models for the detection of GPS spoofing	Custom dataset	GBDT, RF, LightGBM, XGBoost	XGBoost: Accuracy 95.52%, detection time 2 ms, memory size 0.01 MiB	+ Methodology of dataset generation + Taking into consideration memory size and processing time - No hyperparameter tuning
Jullian et al. (2021)	Comparing the performance of MLP & LSTM to be integrated in an GPS spoofing detection framework	UAV ATTACK, TEXBAT	MLP, LSTM	LSTM: Accuracy 99.93% & F1 92.31% in UAV ATTACK, MLP: Accuracy 83.23%, F1 82.79% in TEXBAT	+ Good preprocessing + Adoption of two different datasets - Models complexity difficult to be implemented on board a UAV
Greco et al. (2021)	Framework to detect reactive jamming against UAS using RSSI, throughput & PDR	Vanetjamming 2014 (Puñal et al. 2022) + Simulated data	DT, MLP	MLP: Accuracy 96%, detection time 0.4 s, memory size 17 KB	+ Implementation on RPi to evaluate execution time & memory occupation + Study of generalizability - Adoption of VANET dataset
Tao et al. (2021)	Deep Reinforcement Learning framework to detect intrusions in UAV networks with potential applications	Simulated data	DDPG	Detection ratio greater than 90%	+ Generalizability of the solution over several UAV platforms - Insufficient evaluation metrics & no hardware implementation
DC Bertoli et al. (2021)	Detection of Dos attacks targeting a UAS with consideration of deauthentication, TCP, and UDP flood scenarios	Custom dataset	LR, DT	DT: Accuracy F1 97%, detection time 6.75 ms%	+ Consideration of several techniques of DoS attacks in the dataset + Explainability of the model - Model not evaluated on-board a UAV
Gaber et al. (2021)	Enhancing the performance of a previous study through data cleaning	Dataset from study (DC Bertoli et al. 2021)	LR, DT, KNN, RF	DT: F1 98.9%, runtime 3.29 ms	+ Studying the impact of data cleaning on models' performance - No deployment
Titouna and Naït-Abdesselam (2021)	Detection of FDI based on mobile agents that collects data for attack detection	Simulated data	ANN	DR 94%	+ Comprehensive framework including several components - Insufficient evaluation metrics
Mehmood et al. (2022)	Detection network attacks through multiple components, namely parser, selector, router and performance analyzer	CSE-CIC-IDS 2018	RF, DT, KNN	RF: Accuracy 95%	+ Discussion of the correlation between entropy in CPU utilization, network traffic, and the existence of attacks - Unique evaluation metric

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Whelan et al. (2022)	MAVIDS, a comprehensive sensor attacks detection system that includes a GCS web application, an on-board IDS agent, and a customized communication protocol	UAV ATTACK	AE, OC-SVM, LOF	AE: F1 90.57% and 94.3% in detecting GPS spoofing and jamming attacks respectively, prediction time 0.417 ms	+ Creation of a real-world dataset + Generalizability across UAV platforms + Hardware implementation of the solution - Inability to distinguish attacks from UAV anomalies
Baig et al. (2022)	Detection of hijacking, GPS signal jamming, and DoS attacks	Reverse-engineered UAV Log Data	RF, NB, LR, SVM	RF: Accuracy 97.84%, training time 0.2544	+ Reproducibility of results verified - Dataset based on assumptions made on flight logs not on real cyberattacks
Wei et al. (2022)	ConstDet: detection of GPS spoofing based on analyzing the control process consisting of horizontal position and altitude controls	Custom dataset	SVM, KNN, RF, DT, GBDT, MLP, XGBoost	XGBoost: Accuracy 97.7%, F1 97.72, AUC of 99.7%	+ Real-world experiments + Models deployed on a real UAV - No hyperparameters tuning
Wei et al. (2022)	PerDet: Detection of GPS spoofing based on perception data, such as accelerometer, gyroscope, and GPS	Custom dataset	SVM, KNN, RF, GBDT, XGBoost	XGBoost, RF: Accuracy 99.7%, F1 99.22%	+ Framework compared to other approaches + Real experimental flights & scenarios - No computational overhead evaluation
Sung et al. (2022)	A lightweight 1D CNN model trained on SNR features to detect attacks quickly with less resources	Custom dataset	1D CNN ResNet, SVM	1D CNN: detection time below 50 ms, power consumption below 5 W in embedded systems	+ Implementation on embedded boards and on a real UAV + Computational overhead evaluation - Not enough details about the dataset
Gasimova et al. (2022)	Comparison of the ensemble learners bagging, stacking, boosting for GPS spoofing detection	Dataset from study (Aissou et al. 2021)	GBDT, KNN, NB, DT, RF, LR	Stacking: Accuracy 95.4%, memory size 191 MB, detection time 0.24 s	+ Different configuration of ensemble learners - Memory size relatively high
Aissou et al. (2022b)	Comparison of instance-based supervised ML models for GPS spoofing detection	Dataset from study (Aissou et al. 2021)	SVM, C-SVM, nu-SVM, KNN, RN	nu-SVM: Accuracy 92.78%, detection time 0.12 s, memory size 0.508 MiB	+ Diversity of evaluation metrics - Probability of misdetection relatively high
Khoei et al. (2022)	Comparing unsupervised ML to supervised ML models in terms of accuracy, training time, prediction time, and memory size, among others	Dataset from study (Aissou et al. 2021)	K-means, PCA, AE, NB, LR, ANN, RF, CART, SVM	AE: Accuracy 99.53%, prediction time of 0.39 s, memory size 150.2 MiB, CART: prediction time 0.11 s, memory size 142.6 MiB	+ Comprehensive performance evaluation with eight metrics - Preprocessing & hyperparameter tuning not well detailed
Viana et al. (2022a)	DL approach for detection of jamming in 5 G communication based on signal received by UAVs	Simulated data	CNN-LSTM	Accuracy 99%	+ Comparison of proposed approach to statistical method - Lack of computational overhead evaluation and execution time
Viana et al. (2022b)	Detection of jamming in 5 G communication based on SINR and RSSI of OFDMA received signals	Simulated data	CNN-LSTM, CNN-Attention	CNN-Attention: 74% of the samples correctly identified	+ Variety of simulated scenarios with attacker moving/static - Adoption of insufficient evaluation metrics

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Zhang et al. (2022)	UAS multifractal analysis IDS to detect communication interception by combining WLM & ML	dataset from RADAR records	Bi-LSTM	Accuracy 90%	+ Generalizability as the approach is independent of the UAV platform - Limited performance evaluation metrics
Li et al. (2022a)	Detection & Classification of jamming attacks in OFDM-Based UAVs via Feature- & Spectrogram-Tailored ML	UAVS jamming dataset (Li et al. 202b)	CNN, DT, KNN, LR, MLP, NB, RF	CNN EfficientNet-B0 in spectrogram approach: Accuracy 99.20%, FAR 0.03%	+ Adoption of two separate approaches + Creation of a real-world dataset - Limited performance evaluation
Price et al. (2022)	Detection & classification of data link attacks into barrage, single-tone, successive-pulse, protocol-aware attacks	UAVS jamming dataset	RF	Detection rate 93%, detection time 1.5 ms	+ Implementation on a real UAS - Only one model evaluated - Additional hardware set-up
Slimane et al. (2022b)	Lightweight ML models for detection of deceptive jamming attacks	Simulated data	XGBoost, GBDT, LightGBM	LightGBM: Accuracy 98%, detection time 0.041 s, memory size 0.016 MiB	+ Broad range of evaluation metrics + Well detailed preprocessing - No hardware implementation
Ouiazzane et al. (2022)	Multi-agent ML-based NIDS to detect Dos attacks in a fleet of drones	CIC-IDS 2017	KNN, RF	DT: Accuracy 100%	+ Well elaborated framework - No experiment with real UAS
Da Silva et al. (2022)	Detecting DoS attacks on a wireless UAV network as part of an MQTT-based platform	AWID2	XGBoost, CatB, LightGBM	LightGBM: TPR 97%, memory usage 36 MB	+ Implementation on DJI Tello UAV/RPi - Use of a dataset not specific to UAS
Naït-Abdesselam et al. (2022)	Detecting FDI in the GCS by comparing images received from adjacent UAVs using CNN model	VisDrone dataset	CNN	Accuracy 99.2%	+ Lightweight compression of images on-board the UAV - Detection possible only for drones flying near each other
Slimane et al. (2022a)	Detection of ADS-B attacks, such as path modification, ghost aircraft injection and velocity drift	Simulated data	Linear-SVM, c-SVM, nu-SVM	c-SVM: Accuracy 95.32%, detection time 0.62 s, memory usage 0.0391 MiB	+ Several attack scenario in the created dataset - Assessing solely SVM models
Abu Al-Haija and Al Badawi (2022)	CNN-based IDS that uses a shallow network architecture to detect attacks in UAS networks	UAVIDS-2020	CNN	Accuracy 99.5%, prediction time 2.77 ms	+ Comparing the solution to other approaches + Computational overhead evaluation - Use of unsuitable dataset
Wu et al. (2023)	Interpretable CNN-BiLSTM-Attention model for attack detection using real-time UAV sensor	Custom dataset	CNN-BiLSTM-Attention	Detection rate 99.4%, prediction time 367 μs	+ Identification of the most valuable features for each type of attacks through SHAP interpretability - No implementation of the model
Tlili et al. (2023)	Dynamic intrusion detection framework (DIDF) that detects attacks in UAVCAN message exchanges	UAVCAN (Kim et al. 2022)	LSTM	Accuracy 99.73%	+ Proposed approach compared to other UAVCAN detection methodologies - Limited number of evaluation metrics

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Wu et al. (2023)	Q-learning-based cooperative intrusion detection system through dynamic voting & auditing	Simulated data	Q-learning	Accuracy approaching 99%, lowest energy consumption	+ Comparison to other approaches + Evaluation of power consumption - No implementation of the proposed solution
Tocci et al. (2023)	DQN RL approach to detect intrusions in UAVs	CIC-IDS2017	Deep-Q Network	Accuracy above 90%, prediction time 51.3 FPS	+ Solution deployed on FPGA boards - No real implementation of the solution
Schermann et al. (2023)	Creating and implementing a network DL-based IDS for UAS by analyzing network traffic	CSE-CIC-IDS 2018, IoT-Dataset-2019	MLP, RNN, LSTM, SVM, NB	MLP: Accuracy 96.44%, prediction time 4.42 s	+ Implementation on a embedded system - Prediction time relatively high - Used datasets not specific to UAS
Sajid et al. (2023)	ML-based NIDS integrated into a fog-based framework to classify UAV behavior as normal or malicious	CIC-IDS 2017	XGBoost, Extra Trees, RF, DT, K-means	XGBoost: Accuracy 99.7%	+ Modular design - Evaluation of CPU and memory usage - The framework needs to be adapted to other applications
Ntizikira et al. (2023)	CNN-LSTM for real-time network anomaly detection and precise threat identification	CIC-IDS 2017	CNN-LSTM	Accuracy 99.98%, F1 99.92%	+ Approach compared to state-of-the-art methods - Problem of computational requirements, scalability, and generalizability
Khalil and Rahman (2023)	ANFIS that combines ANN and fuzzy deduction frameworks to detect sensor attacks	UAV ATTACK	ANN, DT, NB, LR, LDA, SVM, TSK	ANFIS: F1 around 85%, memory size 5 KB, execution time 3024 μs	+ Approach implemented on a microcontroller + Lightweightness of models evaluated - Approach needs to be adjusted to other UAV platforms
Da Silva et al. (2023)	Separately detecting attacks targeting the UAV GPS receiver and the UAS network through two subsystems	UAV ATTACK, WSN-DS	AE, OC-SVM, LOF, IF, LightGBM	AE: F1 92.9%, LightGBM: F1 97.9%	+ Handling data imbalance problem + Detecting a wider range of attacks - Lack of evaluation metrics, such as memory usage and detection time
Sun et al. (2023)	PCA for flight control feature reduction, CNN for feature selection, LSTM to model temporal dependencies for detection of GPS spoofing	Custom dataset	Several ML and DL models	PCA-CNN-LSTM: F1 97.22%, Accuracy 99.49%	+ Comprehensive comparative analysis - No computational overhead evaluation - No hardware implementation
Dang et al. (2023)	Comparing GPS topology information with communication topology in cellular-connected UAV swarm to detect GPS spoofing	Simulated data	GNN	GNN: Accuracy 90%, detection time below 10 ms	+ Low detection time + Considering random wind noise - Limited number of evaluation metrics
Brewington and Kar (2023)	Exploring Neural Generative models and evaluating their performance compared to traditional one-class models	Custom dataset	AE, IF, LOF, OC-SVM, Bi-GAN	Bi-GAN: Accuracy 87.6%, F1 82.7%, memory size 56 KB, prediction time 16.3 ms	+ Efficiency metrics for deployment feasibility + Comparative analysis - Approach not tested on real UAS

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Nayfeh et al. (2023)	Real-time detection based on MAVLINK messages forwarded directly from the UAV flight controller to its flight computer	GPS spoofing dataset (Nayfeh et al. 2023)	DT, RF, KNN, MLP, NB, SVM	RF: Accuracy 91.78%, detection time 43.36 ms	+ Implementation in RPi connected to a UAV + Short detection time - Approach limited to UAV applications with predefined paths
Nayfeh et al. (2023)	Comparing the performance of ML models on location-dependent and location-independent datasets	GPS spoofing dataset (Nayfeh et al. 2023)	RF, KNN, MLP, LR, DT, SVM, NB	DT: recall 92.36%, LR: recall 96.67% detection time 0.024 μs	+ Creation of two datasets: location-dependent & location-independent - Lack of computational overhead evaluation
Viana et al. (2023)	Deep Attention Recognition solution for detecting jamming attacks	Simulated data	DNN-LSTM, RF, DNN-Att, CatB, XGBoost	DNN-Attention: Accuracy 90.8%, detection time 30.8 ms	+ Consideration of LOS, BLOS, and LOS/BLOS jamming scenarios + Comparative analysis - No performance overhead evaluation
Burns et al. (2023)	Comparison of hybrid models for the detection of GPS spoofing and their resistance to Gaussian noise	UAV ATTACK + simulated data	PCA-SVM, AE-LR, LSTM-LR	PCA-SVM: Accuracy 99.96%	+ Testing the effectiveness of the proposed models under perturbations - Absence of memory size evaluation & details of simulated data
Khoei et al. (2023)	Comparing performance of supervised DL models to detect GPS spoofing in terms of accuracy, prediction time, and memory size, among others	Dataset from study (Aissou et al. 2021)	LSTM, DNN, U-Net	U-net: Accuracy 98.8%, detection time 0.2 s, memory size 199.87 MiB	+ Adoption of timing and memory metrics + Comprehensive comparison - Proposed models difficult to implement due to complexity
DC Bertoli et al. (2023)	Autonomous ML detection of malicious traffic in the UAV-GCS link through MAVLINK packets	Custom dataset	DT	F1: 99%	+ Well-detailed testbed - Limited performance evaluation metrics - Unique ML model
El alami et al. (2023)	Combining NB & MLP to detect GPS jamming attacks performed by several jammers against a UAV	Simulated data	SVM, KNN, XGBoost, NB, MLP	NB-MLP: Accuracy 96.5%, F1 96.5%, detection time 216 ms	+ Investigating the effect of increasing the number of jammers on performance - No computational overhead evaluation
Chulerttiyawong and Jamalipour (2023)	Sybil attack detection based on two monitoring nodes positioned on the ground, utilizing the features of radio signals for location verification		DT models	DT (J48): Accuracy 91%	+ Solution less susceptible to attack on upper layers - Requires a permanent link between the UAV and the two nodes
Aladi and Alsusa (2023)	Verifying the legitimacy of received signals through geographic coordinates, RSSI, and AoA characteristics, to detect identity spoofing	Simulated data	several DL models	AE: F1 98.75%	+ High accuracy - No hardware implementation - No computational overhead evaluation

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Hassler et al. (2024)	Fusing cyber & physical features to detect several attacks in UAS	Custom dataset	SVM, FNN, RNN-LSTM, 1D-CNN	1D-CNN: Accuracy 94.52%, F1 96.13%	+ Improved IDS generalization ability against unseen attacks + Interpretability of results - No computational overhead evaluation
Hadi et al. (2024)	Fusion multi-tier DNN model for critical 6 G networks enabled by UAV	UAVIDS-2020, NF-UQ-NIDS2-v2, 5 G-NIDD	MLP, CNN, GAN	Accuracy 99.25, detection time 118 ms	+ Implementation on real UAS - Use of unsuitable dataset
Ihekoronye et al. (2024)	DroneGuard: an explainable ML framework to detect GPS spoofing and DoS attacks	Dataset from (Aissou et al. 2021), WSN-DS	RF, DT, LR, GNB, AdaBoost, XGBoost	DT: Accuracy above 94.22%	+ Comprehensive evaluation + Comparison with other approaches - Use of datasets not specific to UAS
Wu et al. (2024)	Tiny IDS based on SDL model with fuzzy rough set as a feature selection technique	CIC-DDOS2019, CSE-CIC-IDS 2018	RF, SDL, SVM, CNN, GAN, LSTM	SDL+RF: Accuracy 99.51%, F1 99.72%	+ Ablation study to confirm the necessity of each component of the proposed approach - No implementation of the solution
Bayrak (2024)	Detecting anomalies in UAV communication traffic using linear-SVM	UAVIDS-2020	Linear-SVM	Accuracy above 99.94%	+ Model interpretability through LIME technique - Use of unsuitable dataset
Hong and Yoo (2024)	Multiple models running simultaneously to detect flooding, fuzzy, and replay attacks	UAVCAN (Kim et al. 2022)	LSTM, DT, LR, KNN, RF, DNN, RNN	F1 97%	+ Model interpretability through SHAP - No computational overhead evaluation or hardware implementation
Tufekci et al. (2024)	Detection of TCP & UDP flooding, deauthentication and MITM attacks using supervised ML	Custom dataset	GBDT, linear-SVM, DT, KNN, RF, Extra Trees	GBDT: Accuracy 95.84%, memory usage 80 MB, detection time 9.86 μs	+ Fast detection time + Comprehensive evaluation - No preprocessing and hyperparameter tuning
Zeng et al. (2024)	GAN to build synthetic datasets integrated with Human-in-the-Loop to strengthen the decisions of ML models for the detection of network attacks	Synthetic data from CIC-IDS 2017 dataset	GAN, RF, KNN, XGBoost, DT, SVM	XGBoost: Accuracy 99.79%	+ Novel approach with high accuracy - Approach resource intensive with respect to human factor - No computational overhead evaluation
Alzahrani (2024)	Multi-sensor anomaly detection approach by investigating the effect of fusing sensor data on performance	UAV ATTACK	SVM	Accuracy 99%, AUC 1	+ High accuracy - Unique ML model evaluated - Insufficient evaluation metrics - Lack of computational overhead evaluation
Alva et al. (2024)	Detecting GPS-based attacks by learning novelties in PWM signals	Custom dataset	IF, OC-SVM, LOF	LOF: Accuracy 75.87%	+ Generalizability over other UAV platforms - Low accuracy - Insufficient evaluation metrics
Miao et al. (2024)	CNN-BiLSTM model optimized by GG to detect sensor-based attacks	Simulated data	GG, CNN BiLSTM Attention, NB, SVM	CNN BiLSTM Attention: Accuracy 98%, prediction time 201.23 s	+ Relatively high accuracy - No hardware implementation - High execution time - Attack scenarios not well-detailed

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Mughal et al. (2024b)	Leveraging spatial and temporal communication patterns to detect cyberattacks	Custom dataset	GNN, FNN, LSTM, and 1D-CNN	GNN: Accuracy 98.91%	+ High accuracy + Well-detailed testbed - No computational overhead evaluation
Hadi et al. (2024)	An autonomous collaborative IDS framework with DL detection and incident response handling	UAVIDS-2020	CNN	Accuracy 98.23%	+ Adoption of a real-world testbed + Comparison to other approaches - Insufficient evaluation metrics - Use of unsuitable dataset
Wei et al. (2024)	SigFeaDet: Detection of attacks by recognizing anomalies in CN0 and Doppler frequency characteristics	TEXBAT+ custom dataset	DT, RF, GBDT, XGBoost, KNN, SVM	KNN: Accuracy 95.56% F1 95.98%, prediction time less than 0.5 ms, 55% increase in CPU usage	+ Implementation on real UAV + Computational overhead evaluation - Additional hardware - Solution tested only on benign samples
Korium et al. (2024)	Signal-based dataset transformed into RGB images before being trained on CNN models to detect GPS spoofing	UAV ATTACK (Whelan et al. 2020)	CNN models	CNN-VGG-19 on unseen dataset: F1 99.249%, detection time 2.72 s	+ Use of real-world dataset + High generalization capability - No hardware implementation - No computational overhead evaluation
Wei et al. (2024)	An interpretable framework, Trust AI-based Decentralized Anomaly Detection (TADAD), based on MAVLINK UAV logs	Custom dataset	MLP, RF, AdaBoost, GBDT	MLP: F1 and accuracy 98.6%	+ Interpretation of the results by SHAP + Multiple GPS spoofing scenarios - No hardware implementation - Limited evaluation metrics
El alami and Rawat (2024)	DroneDefGANt approach that combines the GAN and transformer models to detect GPS spoofing attacks	Simulated data	GAN, KNN, LSTM, CNN and SVM	GAN: Accuracy 97%, F1 97%	+ Performances not affected by Gaussian noise - No hardware implementation or evaluation in real environment
Alhoraibi et al. (2024)	Comparing DL models performance when trained with and without adversarial samples built by a GAN model	Custom dataset	CNN and LSTM	Training with adversarial samples increased F1 by 12%, GAN+CNN: Accuracy 98%	+ Resilience of the detection mechanism against adversarial attacks - No hardware implementation or overhead evaluation
Ferrão et al. (2024)	Comparison of eight supervised models for the detection of GPS spoofing	UAV ATTACK	LightGBM, XGBoost, RF, DT, SVM, GBDT, MLP	RF: Accuracy 98%, execution time 1.16 s	+ High accuracy and F1 score - Lack of evaluation of computational overhead
Finn et al. (2024)	Comparing ML to DL models to detect stealthy UAV GPS spoofing attacks by fusing sensor data	Custom dataset	XGBoost, GBDT, LSTM, bi-LSTM, RNN	XGBoost: Accuracy 98.7%, AUC 99.87% in straight flight	+ Diverse flight scenarios in the newly created dataset - Absence of computational overhead evaluation and detection time
Ghelani et al. (2024)	Gradient Monitored Technique Based on Reinforcement Learning (RLGM) for identifying jammers within a UAV network	Simulated data	RLGM, Federated RL, DQL	Average Accuracy 80%, running time 7.5 s, less than 80 joules	+ Evaluation against other approaches in terms of power consumption, running time and accuracy - No hardware implementation

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Sharma et al. (2024)	Detecting TCP and ICMP flooding through TST	Custom dataset	XGBoost, IF, LSTM, BiLSTM	TST: Accuracy above 97%, memory usage 371.32 MiB, detection time 133.9 ms	+ Creation of real-world dataset + A comprehensive evaluation - Memory usage relatively high
Alkhatib et al. (2024)	Classifying GPS jamming into four types & indicating the direction/the distance between the UAV and the jammer	Custom dataset	RF, KNN, MLP, LR, DT, NB, SVM	MLP: F1 98.9%, prediction time/sample 50 μs	+ Creation of an enriched dataset with different attack scenarios - No memory size occupation evaluation or hardware implementation
Almeida et al. (2024)	Side-channel ML approach based on changes in impedance, reactance, and resistance to detect HT in UAV control circuits	Custom dataset	RF, LR, KNN, SVM, XGBoost	XGBoost: Accuracy 98%, detection time 86 ms, memory size 197 MB	+ Discussion of the practical implementation of the proposed approach + Well-detailed HT attack scenario - Additional hardware
Renu and Sarveshwaran (2024)	Detection of Wormhole & sinkhole based on round trip time, PDR, and signal strength, among others	Simulated data	Sheppard CNN & Spinal Network	ShCSpinalNet: DR 93%	+ Comparative analysis and discussion - No hardware implementation
Luo and Oracevic (2024)	Identifying anomalies resulting from environmental influences on UAVs, cyberattacks, and specific sensor malfunctions	Custom dataset	Bi-LSTM, single LSTM, Bi-LSTM with attention	Accuracy of 99.89% for attack detection, 96.83% for windy detection	+ Diversified simulated flight scenarios and conditions - High computational complexity
Ma et al. (2024)	Stacked ensemble learning that combines CNN for feature extraction, XGBoost as base learner, and LR for final prediction of GPS spoofing	UAV ATTACK	DT, GBDT, SVM, KNN, MLP, XGBoost, CNN-XGBoost	CNN-XGBoost: Accuracy of 99.79%	+ High accuracy - The implementation of feature extraction with CNN is not sufficiently explained
Liu et al. (2024)	GPS spoofing detection on UAVs through image matching technique	Custom dataset	Siamese ResNet and SuperGlue	88.4% in accuracy, detection time 100 ms	+ Generalization test - shortcoming in complex atmospheric conditions and during night
Aldossary et al. (2025)	Cross-Layer Convolutional Attention Network to detect attacks for Drone-to-Drone (D2D) and Drone-to-Base Station (D2BS) communications	Dataset from (Keiko 2024)	CLCAN, CNN, LSTM, XGBoost	CLCAN: Accuracy 98.4%, F1 98.7%	+ Optimized computational efficiency + Lowest time complexity - No computational overhead evaluation
Mouatassim et al. (2025)	A single predictive model to identify the occurrence of GPS jamming, as well as its direction and distance	Dataset from (Alkhatib et al. 2024)	DT, XGBoost, KNN, LR, NB	XGBoost: F1 99.95%, prediction time : 96.62 ms	+ Good preprocessing - No hardware implementation - No computational overhead evaluation
Chen et al. (2025)	Detecting attacks by training unsupervised models on a multi-attack dataset	Custom dataset	IF, EE, and LOF	IF: Accuracy 99.84%	+ Creation of a comprehensive dataset + High accuracy with minimal influence by noise and drift - Insufficient evaluation metrics

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
Du et al. (2025)	UGL Network attack detection model specifically designed for UAV controller by combining GCN and LSTM	UAVCAN dataset	GCN, LSTM, ANN, KNN, NB, SVM, DT	UGL: Accuracy of 100% for flooding, 98.54% for fuzzy attacks, and 96.35% for replay attacks	+ Broad comparative analysis - Insufficient evaluation
Hakeem et al. (2025)	MobileNet, Transfer learning and deep learning models for GPS spoofing attack detection	Dataset (Aisso et al. 2022a)	CNN, MLP, LSTM, ResNET, Inception, among others	MobileNet: Accuracy of 98.49%, F1 99.20%	+ High accuracy after feature selection - No consideration of time or discussion of implementation
Al-Syouf et al. (2025)	GBM-RFE-based ensemble models for the detection of GPS spoofing	Datasets (Nayfeh and Shamaileh 2022; Aisso et al. 2022a)	DT, RF, KNN, MLP, DNN	Bagging classifier: Accuracy higher than 99.12%, processing time lower than 29 ms	+ The use of two real-world UAV relevant benchmark datasets - No computational overhead evaluation
Islam et al. (2025)	DRL to detect attacks against UAVCAN networks	UAVCAN dataset	Q-Network	Accuracy 96%	+ Adaptability to changing environment - The study lacks comparative analysis and extensive performance evaluation
Al-Sabbagh et al. (2025)	LSTM networks optimized by GA to detect GPS spoofing	Dataset (Aisso et al. 2022a)	LSTM, LSTM-GA	LSTM-GA: Accuracy 93.12%, F1 93.39%	+ Comprehensive evaluation + Comparative analysis - High computational complexity
Talaei Khoei et al. (2025)	Capsule networks for detecting GPS spoofing attacks on UAS	Dataset (Aisso et al. 2022a)	CapsNet architectures, CNN	Efficient-CapsNet: Accuracy of 99.1%, prediction time of 0.5 s, memory size 123 Mb	+ Comparative analysis with other relevant works + Comprehensive evaluation - No computational overhead evaluation
Yu et al. (2025)	Detecting attack through an end-to-end multi-task RL framework enhanced by spatiotemporal attention, causal inference, and action prediction	Simulated data	MTD3QN and other models	MTD3QN: F1 99.30%	+ High performance in various environments - High computational complexity - Idealized environment
Badar et al. (2025)	DeepSpoofNet using ANOVA-selected features and CNN and BiLSTM layers for GPS spoofing detection	UAV ATTACK	ANN, CNN, CNN-BiLSTM	CNN-BiLSTM: 99.25% in accuracy	+ Good preprocessing - Insufficient evaluation metrics
Wei et al. (2025)	One single ML model to classify various sensor attacks using heterogeneous sensor data and control parameters	Custom dataset	MLP, KNN, RF, GBDT, XGBoost	MLP: online detection rate of 74%, 0.495 ms per detection, model size 48 KB	+ Improved computational efficiency + lightweightness of models - Sensor attacks simulated through software injection

Table 6 (continued)

Ref	Approach	Dataset	Models	Results	Strengths & limitations
She et al. (2025)	Improved AdaBoost-CNN model trained with limited signal features-based samples	Simulated data + TEXBAT dataset	AdaBoost-CNN, CNN, DNN, SVM, KNN	AdaBoost-CNN: Accuracy 95.83%	+ High detection accuracy under low SNR conditions - No evaluation of computational overhead and model's complexity

$$\text{Energy Overhead (\%)} = \frac{E_{\text{with model}} - E_{\text{baseline}}}{E_{\text{baseline}}} \times 100 \quad (7)$$

Furthermore, prediction throughput is a crucial evaluation metric. It refers to the number of inferences that a model can make within a given time frame. This metric is particularly important for attack detection deployed onboard UAVs, as low throughput may lead to processing bottlenecks (Whelan et al. 2022). For consistent comparison across studies, it is preferable to report the detection time per sample, rather than the total detection time influenced by the size of the evaluated dataset.

Detection design

Designing effective attack detection systems for UAS requires strategic approaches to counter cyber threats. This subsection synthesizes the literature insights on detection design.

Attack Detection Methods

Detection systems employ various strategies to identify UAS attacks. A one detector per attack strategy improves precision by tailoring models to specific attack types. For example, Da Silva et al. (2023) proposed an IDS based on two subsystems, one to detect flight anomalies targeting the UAV GPS receiver and the other to detect UAV network attacks. Similarly, Hong and Yoo (2024) presented an approach that consists of two models running simultaneously to detect multiple intrusions, one to detect flooding and fuzzy attacks, and the other to detect replay attacks. In contrast to the one detector per attack strategy, multi-attack detection results in an increased computational burden and poor scalability when faced with multiple simultaneous sensor attacks, as highlighted by Wei et al. (2025), underscoring the trade-off between comprehensiveness and efficiency.

System architectures

IDS architectures provide structural frameworks for UAS security. Ouiazzane et al. (2022) presented a multi-agent ML-based NIDS to detect DoS attacks on drone fleet networks, using a Sniffer Agent to capture traffic, a Match Checker Agent to compare packets against known signatures, and an Alert Manager Agent to notify administrators, addressing UAV mobility and single-point failure risks. Similarly, Hadi et al. (2024) proposed a distributed collaborative IDS framework for UAV networks, including a detection module, an event correlation engine, and an incident response mechanism. Furthermore, Mehmood et al. (2022) proposed an IDS to be implemented in the GCS, featuring a parser for traffic formatting, a GA-based selector for feature extraction, a router directing data to pre-trained ML models, and a performance analyzer for model validation. Meanwhile, Bouhamed et al. (2021) introduced a framework with an IDPS module per UAV with an intelligent global offline training module implemented at a central station where the drone swarm returns to charge and receives periodic updates of their modules. In contrast, the solution proposed by Aladi and Alsusa (2023) includes two monitoring nodes positioned on the ground within the operational zone to detect whether two signals, which appear to originate from different UAVs, actually emanate from the same location to identify UAV identity spoofing.

Emerging approaches

A novel approach was suggested by Fraser et al. (2021). The researchers proposed a digital twin architecture to detect attacks through novelties. It consists of a digital twin for each UAV, modeling normal behavior by collecting and analyzing historical data in the GCS. The GCS trained models are uploaded and programmed in the UAV flight controller before every flight for real-time monitoring and detection of attacks. The authors proposed elaborating a model for each subsystem of the UAV and relying on inputs of human experts along with historic data to help select the relevant features. Human expertise was also used to strengthen the decisions of the ML models in (Zeng et al. 2024), using a Human-In-The-Loop approach. In fact, given a stream of data, when the confidence of the model in classifying a traffic is below a threshold, this traffic is flagged for further review by a human expert. The researchers concluded that the proposed approach improved the performance of all investigated ML models.

While developing optimized ML models to detect attacks in UAS under resource constraints is crucial, holistic approaches that guide the practical implementation of detection solutions are equally critical. Incorporating human expertise into these frameworks strengthens decision-making, providing robust detection of complex cyberattacks.

Deployment and validation

Training ML models for attack detection and evaluating their performance on a CPU or GPU machine is insufficient to determine their effectiveness when operationally deployed on a UAS. For this reason, some researchers turn to embedded systems to evaluate their models, while others deploy their models on real UAS and conduct validation experiments.

Experiments on embedded systems

Some researchers used real hardware-emulated UAV testbeds to assess the lightweightness of their solutions. The execution time, memory occupation, and power consumption were measured to confirm that the proposed models are capable of running on resource-constrained UAVs.

The memory occupation depends on the nature of the ML model. For instance, the ANFIS (Khalil and Rahman 2023) was implemented on an Arduino Nano 33 BLE Sense microcontroller. It occupied 5 KB of memory for attack detection. In contrast, the DT and MLP models proposed by Greco et al. (2021) were deployed on a RPi 3, and respectively occupied less than 10 KB and 17 KB.

The inference time also depends on the processing capabilities of the system, such as the number of available threads, CPU/GPU performance, and memory bandwidth. By deploying their ML models on an RPi, Greco et al. (2021) registered an execution time of 0.4 s. Meanwhile, when using an additional Intel Neural Computing Stick (NCS2) connected to an RPi, the average inference time for the LSTM–based autoencoder was 4.32 ms, while that of the binary LSTM classifier was 3.986 ms (Agyapong et al. 2021).

Implementing DL or RL models on embedded systems may result in power consumption overhead. For example, the DeepSIM detection system (Xue et al. 2020) deployed on-board an RPi resulted in a power consumption overhead of 51.64%. The trade-off between detection time and power consumption was highlighted by Tocci et al. (2023). The authors deployed a DQN RL approach on a Xilinx Zynq UltraSCALE+ ZCU102 FPGA board. Running inference with a single thread results in a 50.25% power consumption overhead, achieving 2599.02 Frames Per Second (FPS). When additional threads are activated to improve inference speed, power consumption reaches its peak, with the system delivering up to 5492.66 FPS.

Experiments on real UAS

The deployment of a ML-based cyberattack detection system on a real UAS for validation ensures the effectiveness of the model under realistic conditions. It exposes the model to real-world variables such as sensor noise and flight dynamics, allowing evaluation of system integration. Several flight experiments and approaches have been adopted by scholars to validate the effectiveness of their models. Researchers implemented their solutions on UAV platforms integrating it into either a flight companion computer, such as a Raspberry Pi or NVIDIA Jetson, or a microcontroller.

Most researchers deployed pre-trained models onboard UAVs. In contrast, online training was proposed by Feng et al. (2020) to improve generalizability. The researchers proposed a GA-XGBoost method to detect GPS spoofing attacks. The method relies on an XGBoost model trained off-board due to resource constraints, using flight logs with parameters tuned by GA, then uploaded to drone and trained again on-board to adapt to different UAV platforms and improve prediction results. Validation experiments were conducted on a Navio2-based UAV. The authors assumed that the drone is not hijacked within several seconds after take-off, allowing sufficient time for training the XGBoost model. This assumption may not consistently apply.

In addition, data acquisition methods for model prediction differ across researchers, impacting detection time. SigFeaDet (Wei et al. 2024) deployed on board a pixhawk-based UAV, captures GPS signals through a USB-connected SDR, namely RTL-SDR, interfaced with the RPi flight computer. Similarly, the detection solution deployed by Price et al. (2022) includes a HackRF SDR connected to the flight computer for the reception of RF transmissions to detect jamming attacks. The average processing time to classify a sample was 1.5 ms. Both solutions involve an additional hardware setup that adds weight to the UAV and generates delays. Meanwhile, the deployment proposed by Nayfeh et al. (2023) enables real-time solution functionality without additional hardware setup. In fact, the GPS receiver in the COEX drone forwards the GPS signals directly to the RPi processor through the flight controller PX4 without any modification of the MAVLink messages. The RF model classified samples to detect GPS spoofing attacks in less than $70\mu s$ per sample. Likewise, ConstDet (Wei et al. 2022) was implemented on a RPi flight computer via the Dronekit-python library to process real-time flight data for GPS spoofing detection. Before being fed to the XGBoost classifier for detection, flight control and sensors data are retrieved at a sampling rate between 5hz and 25hz, then preprocessed by scaling and unifying data units to match training data formats, potentially adding latency to the detection time.

The validation experiments conducted by some scholars fell short of realistic conditions for simulating real-world attacks. The detection system proposed in (Wei et al. 2024) was tested only on authentic GPS signals. Feng et al. (2020) employed an array on the microcontroller to store spoofed GPS signals and configured a timer to activate a hijacked mode, during which the GPS signal processing program retrieves inputs from the array rather than the GPS receiver. In contrast, Sung et al. (2022) conducted flight tests using a spoofer jammer. The effectiveness of this device was initially demonstrated against two DJI drones before being applied to the UAV equipped with the GPS spoofing detection system, which successfully identified the attack.

Some researchers carried out the validation experiment at different speeds, altitudes, and locations, such as urban parks and sports fields (Wei et al. 2024). Changing location during experiments is particularly important, as multi-path environments might deteriorate the detection system performances, as acknowledged by Sung et al. (2022).

Unfortunately, some scholars reported validating their approaches without elaborating on flight experiments. The models proposed in (Hadi et al. 2024, 2024) were deployed on-board three UAVs, namely the PX4 Vision Dev Kit V1.5, DJI Mini SE, and DJI Mini 3 Pro, but no experimental details were provided by the authors. The same applies to Da Silva et al. (2022) who implemented their solution on-board a DJI Tello UAV.

One particularly interesting study was proposed by Whelan et al. (2022). The authors proposed a comprehensive IDS solution that includes an on-board IDS agent for novelty detection, a MAVLink-based customized communication protocol for heartbeat messages exchange, and a Django-developed web application for managing the solution. To build the on-board IDS agent, the researchers evaluated prediction time per sample and prediction throughput of the DL models on-board three companion computers, namely the NVIDIA Jetson Nano, RPi 4, and RPi Zero. The prediction throughput registered by AE implemented on a RPi 4 was 2340 prediction per second. However, this study lacks computational overhead evaluation.

In real-world scenarios, the companion computer might handle mission payload processing, and computational overhead evaluation is of critical importance to ensure efficient performance and reliable operation of UAVs. Only a few scholars evaluated the computational overhead. Ding et al. (2021) deployed a lightweight and transparent LSTM model capable of detecting sensor-based attacks and flight control attacks on board a Crazyflie 2.0 quadcopter. The average runtime and memory size overheads were 2.6% and 0.34%, respectively. The KNN classifier deployed by Wei et al. (2024) increased CPU usage, power consumption and memory usage by 55%, 131.2%, and 130 MB, respectively. The GBDT model deployed by Tufekci et al. (2024) resulted in a memory usage of 80 MB. The 1D-CNN model proposed by Sung et al. (2022) resulted in an increase of 84.52% over idle power consumption when deployed on NVIDIA Jetson AGX Xavier.

Most of the studies surveyed focused on evaluating the performance of ML models on PCs, which is insufficient to assess the effectiveness of cyberattack detection solutions that are to be deployed on-board UAVs. For example, when an ML model is trained on an RPi, the detection time might be up to 15 times longer than on a PC (Nayfeh et al. 2023). As the performance of ML models should be evaluated on a realistic host system, experimental validation and a description of how the detection solution should be implemented are of high importance. Experiments on embedded systems can partially serve to compute computational and power consumption overhead of ML models. In contrast, experiments on real UAS enable not only to compute the whole system overhead but also to conduct validation experiments with test flights.

Limitations and challenges

The reviewed articles share common limitations that affect the efficiency of their proposed solutions. In this section, we summarize the key limitations and challenges of the proposed solutions in the reviewed primary studies, addressing the sub-questions of **RQ2**.

Limitations

The limitations are related to the datasets used, the experiments carried out, the performance evaluation, as well as the implementation methods.

Use of unsuitable datasets

Many research papers used generic datasets, such as CIC-IDS2017 and WSN-DS, designed for computer networks that may be unsuitable to UAS specificity (Tocci et al. 2023; Ouiazzane et al. 2022). Additionally, some authors relied on datasets based solely on normal flight logs without performing or simulating attacks. Other scholars used a benchmark dataset initially created for detecting three types of drones, i.e UAVIDS-2020 (Zhao et al. 2020). Unfortunately, this dataset continues to be mistakenly used by some researchers for developing ML models to detect cyberattacks on UAS.

The use of generic datasets often stems from a lack of a deep understanding of UAS-specific characteristics. In fact, many researchers treat drones as equivalent to general computer systems, overlooking unique aspects like real-time operational constraints, sensor dependencies, and distinct attack vectors (e.g., GPS spoofing, data link jamming). This misconception leads to the reliance on datasets that do not capture UAS-specific dynamics. Consequently, ML models trained on generic datasets are often ineffective in real-world UAS attack detection. These models may fail to detect domain-specific attacks or exhibit poor generalizability due to mismatched feature spaces or unrealistic attack scenario. Furthermore, the use of UAVIDS-2020 in attack detection is the result of a confusion introduced during the naming of the dataset. In fact, the dataset was basically created to differentiate between encrypted Wi-Fi traffic from a UAV and traffic from non-UAV sources for the detection of drone intrusion (Zhao et al. 2020; Abdullayeva and Valikhanli 2023), and does not include any form of cyberattack. As a consequence, the models would not fire alerts in the case of a compromised UAV acting as an attacker and in the presence of typical attacks, since the binary classification model takes decisions based solely on the nature of Wi-Fi traffic.

Limited flight scenarios and conditions

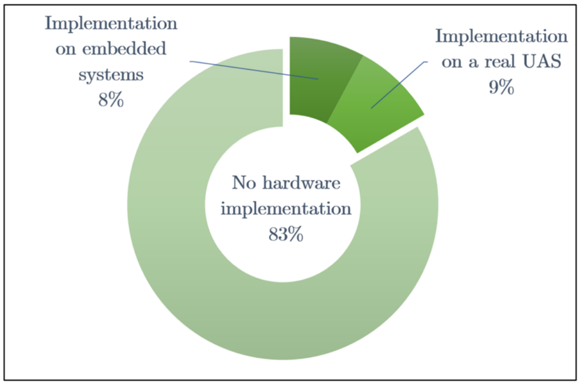

Fig. 11 Distribution of the reviewed studies: Overview of implementation

Only a few scholars conducted experiments with varied flight paths, including circular, square, curved, and random trajectories, while the UAV operated at different altitudes and speeds. Varying fight scenarios during the training phase reduces the risk of overfitting to specific patterns, allowing better generalizability. Some ML models perform poorly on curved and random paths relative to straight-line flights Finn et al. (2024). Furthermore, a diverse set of flight patterns allows the ML model to better differentiate between ordinary anomalies and malicious attacks, reducing false positives when implemented on a real UAS.

Lack of assessment of computational overhead

The lack of a detailed assessment of computational overhead, such as processing time, memory usage, and energy consumption, is one of the most frequent limitations identified in the reviewed articles. During the evaluation of ML models, the majority of scholars focus primarily on performance metrics such as accuracy, precision, recall, and the F1 score, and neglect metrics related to resource consumption, such as energy and memory overhead.

Lack of hardware implementation of ML models

Fig. 11 provides an overview of deployment and validation in the reviewed papers. The figure shows that 83% of the reviewed studies did not include a hardware implementation of the proposed ML models. The remaining papers are evenly divided between deployments on real UAS and deployments on embedded systems. This limitation raises concerns about the reliability, performance and applicability of the ML model in real-world scenarios.

Challenges and future directions

Despite significant advances, ML-based approaches for detecting attacks on UAS face several challenges. These challenges arise from factors such as limited real-world datasets, poor generalizability across UAV platforms, the interpretability of ML models, and high computational costs. Furthermore, the robustness of ML models is often compromised by adversarial attacks, further limiting their reliability in security-critical applications.

Resource efficiency of ML models

On-board solutions are preferred over ground-based ones for real-time attack detection due to potential transmission delays or loss of link with the GCS. However, a major challenge arises during their deployment, particularly concerning computational complexity. This becomes a significant obstacle when implementing ML-based IDS, as UAVs are constrained by limited processing power, memory size, and battery autonomy. Many reviewed studies focus solely on detection performance without considering UAVs as CPS, leading to solutions that are computationally intensive and energy-consuming. These solutions may be impractical for real-world use. Accordingly, the research community should recognize that any solution which is not lightweight and computationally efficient will not be suitable for deployment on resource-constrained UAVs. In fact, UAV IDS solutions should employ lightweight ML models, with evaluation studies assessing computational and energy overhead to ensure resource efficiency. Lightweight models require low computational complexity, measured primarily by floating-point operations (FLOPs)[4](Chen et al. 2023; Chaudhari et al. 2025). Minimizing FLOPs is critical to enable deployment on embedded devices in UAVs (Agiollo and Omicini 2021). To enhance computational efficiency, techniques like pruning and quantization are highly effective (Tocci et al. 2023).

However, to the best of our knowledge, there are no precise standards for lightweight ML models tailored for UAVs; this is due to the diversity of hardware devices used in these vehicles (Ahmed and Jenihhin 2022). A future direction would be to gather researchers and industry to establish clear standards for computational complexity and evaluation metrics, such as energy and memory overhead, to guide the development of lightweight ML models, ensuring their suitability for deployment in UAVs.

Generalizability across UAV platforms

Reviewed studies have shown success in applying ML methods to detection of attacks on UAS, yet these methods are restricted in their effectiveness to particular platforms or specific environments and scenarios. These models are typically trained on data from particular

[4] Libraries like ptflops (PyTorch) or tf.profiler(TensorFlow) can estimate FLOPs for neural networks.

sensors, UAV configurations, or operational scenarios, reducing their ability to generalize across different platforms or environmental conditions. As a result, ML methods may perform poorly when applied to new or diverse UAS, hindering their scalability and robustness in real-world applications. A solution to the generalizability problem is the novelty detection approach, as it does not depend on any specific sensors, UAV platforms, or communication protocols. However, a major limitation of novelty detection is its inability to differentiate between an ordinary anomaly and a deliberate malicious attack.

Resilience against adversarial attacks

One of the main challenges in using ML to detect attacks on UAS is ensuring resilience against adversarial attacks, where malicious inputs are designed to deceive detection systems. These attacks can significantly undermine the reliability of intrusion detection mechanisms. AML has emerged as an effective approach to counter these vulnerabilities, helping to strengthen the system against deceptive inputs. To further enhance resilience, some researchers have turned to Generative AI, using it to create adversarial samples for training. This adversarial training process allows for the development of a generative dataset which, in turn, improves the robustness of the detection mechanism.

Dataset Scarcity

The scarcity of real-world datasets presents a significant challenge in developing robust IDS for UAS. Many datasets remain inaccessible due to institutional policies, with some studies withholding data from public release. Furthermore, while more scholars are creating custom datasets tailored to their specific needs, these datasets are often not shared with the scientific community. In cases where they are shared, they frequently lack proper naming and standardization, limiting their reusability and hindering comparative research. The acquisition and distribution of over-the-air attack datasets is further complicated by privacy concerns, proprietary data ownership, and restrictive regulations in data collection. To overcome these challenges, GANs offer a potential solution by synthetically generating realistic intrusion data, helping to expand available datasets while preserving privacy. However, ensuring that synthetic datasets are accurately labeled and reflect real-world complexities remains crucial for their effective use.

Explainability

Ensuring the explainability and interpretability of ML models is a significant difficulty when employing them to identify intrusions on UAS. Although these models are capable of accurately detecting attacks, it is still challenging to comprehend how and why they generate particular predictions. In order to overcome this difficulty, researchers used methods like LIME and SHAP, which are designed to provide insight into how much each attribute influences the ultimate choice. Even with the application of these techniques, it is still challenging to provide sufficient transparency in intricate DL models for UAS attack detection, since it requires striking the correct balance between model accuracy and findings that are intelligible to cybersecurity experts.

Conclusion

The use of drone systems has expanded across all industries and daily life. Although these sophisticated systems meet various functional needs, they remain poorly secured against malicious attacks from adversaries, especially as hacking tools become increasingly accessible to both trained and untrained individuals. Detecting these attacks in a timely manner remains one of the critical challenges faced by the drone industry and academia. In response, significant attention has been devoted over the past six years to addressing this challenge using ML techniques.

Unlike attack detection in computer systems, detecting attacks against UAS requires considering both their cyber and physical dimensions, as these systems operate in dynamic environments susceptible to several threats such as GPS spoofing, sensor tampering, or data link jamming. To address this complexity, researchers have developed ML models that take advantage of a combination or a specific type of feature data, namely flight control and sensor data, internal circuitry properties, external signal characteristics, network traffic characteristics, and payload data **(RQ1(a))**. Given the often redundant and intercorrelated nature of these feature data, scholars have applied advanced preprocessing techniques, including data cleaning to sanitize datasets and feature engineering to extract attributes and transform data. Also, researchers have leveraged feature selection to reduce dimensionality through various techniques, such as PCA, RFE, correlation analysis, GA and FRS. Additionally, they have adopted dataset balancing to address class imbalances in attack scenarios using SMOTE techniques and their variants, ADASYN, GAR and GANs **(RQ1(b))**. Regarding the learning phase, researchers have employed various paradigms, mainly supervised learning for labeled attack datasets, unsupervised learning through novelty detection to identify unseen attacks, reinforcement learning to adapt to evolving threats in UAS operations, and adversarial learning to resist manipulated inputs **(RQ1 (c))**. To evaluate their approaches, most of the researchers relied on standard metrics such as accuracy, precision, recall, and F1 score, while only a few incorporated measures of energy and computational overhead, inference time, as well as prediction throughput **(RQ1(d))**. Furthermore, some researchers have deployed ML solutions on UAV platforms, integrating them into companion computers such as Raspberry Pi or NVIDIA Jetson or microcontrollers, while others evaluated their approaches on embedded systems, measuring execution time, memory usage, and power consumption to verify suitability for resource-constrained UAVs **(RQ1(e))**.

The primary studies reviewed exhibit shared limitations that may probably impact the effectiveness of the proposed solutions when deployed in real-world, namely the use of unsuitable datasets, limited flight scenarios during training and validation, lack of assessment of computational overhead and hardware implementation **(RQ2(a))**. Although ML-based approaches for detecting attacks on UAS have made substantial progress, they encounter multiple challenges, namely scarcity of real-world datasets, poor generalizability across UAV platforms, high computational costs, vulnerability to adversarial attacks, and explainability **(RQ2 (b))**.

Key academic research priorities for detecting cyberattacks on UAS include advancing edge computing, 6 G network integration, and federated learning. In fact, edge computing is emerging as a research hotspot in several UAS applications, including fire rescue (Sun et al. 2025), agriculture (Ayranci and Erkmen 2024) and object reconnaissance (Tulembayeva et al. 2024). For attack detection, edge computing should be leveraged in future works, as it enables on-board or edge-node inference, minimizing latency and bandwidth demands for real-time detection with limited resources. In addition, integrating UAS with 6 G networks improves autonomous operations and supports novel applications, such as Intelligent Reflecting Surface(IRS)-assisted UAVs to revolutionize wireless communication (Ning et al. 2025). This requires robust security frameworks and new attack detection mechanisms that should be addressed in upcoming research efforts. Furthermore, federated learning is vital for decentralized, privacy-preserving data processing and improving detection accuracy against cyber threats. As it has been applied in swarm anomaly detection and noise removal for IoT UAVs (Lu et al. 2024), taking advantage of multiparty participation and distributed computation, more effort should be devoted to applying this technique to attack detection. Overall, research should focus on ML-based IDS that utilize heterogeneous computing resources, balancing security, throughput, and energy efficiency.

Meanwhile, deploying an ML-based cyberattack detection system on UAVs involves navigating trade-offs between detection performance, such as detection time and accuracy, and resource constraints. In fact, DL and RL approaches improve accuracy but increase memory, power, and computational demands, potentially shortening mission time due to limited battery capacity. Lightweight models are a great solution to manage this trade-off. Another trade-off that should be managed is the data acquisition sampling rates, as higher rates improve real-time detection speed but strain resources, while lower rates risk missing cyberattacks. Finally, validation experiments often lack realistic conditions, undermining reliability, yet comprehensive testing is required to validate each proposed solution.

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